

Antarctic Microbial Sensor

Group 14



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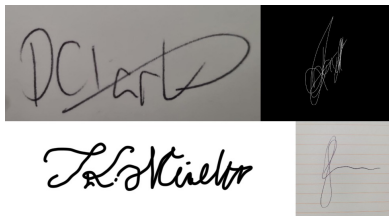
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Abbreviations

ANN Artificial Neural Network

ATP Acceptance Test Procedure

BLE Bluetooth Low Energy

CNN Convolutional Neural Network

CTD Conductivity, Temperature, Depth

DL Deep Learning

DMS Data Management Subsystem

ECR Edge Computing-RESTful

eNVM Embedded Non-Volatile Memory

GUI Graphical User Interface

IDE Integrated Development Environment

IoT Internet of Things

LAMP Loop-Mediated Isothermal Amplification

LED Light Emitting Diode

LoRaWAN Long Range Wide Area Network

LPWAN Low-Power Wide-Area Network

MEMS Micro-Electro-Mechanical Systems

ML Machine Learning

MRAM Magnetoresistive Random Access Memory

NFC Near-Field Communication

NVIS Near Vertical Incidence Skywave

PCB Printed Circuit Board

PCM Phase Change Materials

PDMS Polydimethylsiloxane

ROV Remotely Operated Vehicle

SLA Stereolithography

STT-MRAM Spin-Transfer Torque MRAM

SVM Support Vector Machine

SWPDT Simultaneous Wireless Power and Data Transfer

UI User interface

USB-C Universal Serial Bus Type-C

USV Unmanned Surface Vehicle

Wi-Fi Wireless Fidelity

Chapter 1

Introduction

1.1 Background

Marine microorganisms are the foundational drivers of the ocean ecosystem, mediating global biogeochemical cycles and acting as the first responders to environmental stressors such as climate change and anthropogenic pollution. As Dr Emma Rocke, a marine microbiologist at the Marine and Antarctic Research for Innovation and Sustainability (MARiS) institute at the University of Cape Town, describes it: “*Microbes are the invisible engines of the ocean ecosystem.*” Her research spans the Benguela Upwelling System, the Indian Ocean, and the Antarctic and Southern Ocean, where microbial community dynamics serve as the earliest available proxy for broader ecological shifts.

Gathering the continuous, high-resolution data required to detect these shifts presents a fundamental challenge in the Antarctic region. Current practice relies on research cruises that are expensive, infrequent, and offer only limited spatial coverage. The austral winter exacerbates this shortfall severely: expanding sea ice, temperatures routinely below -30°C , and inclement weather render the region physically inaccessible for months at a time, creating what is widely termed the *winter data gap*. Critically, this period of observational absence coincides with the time at which early detection of harmful algal blooms and ecosystem disturbances would be most valuable. The reliance on human operators to bridge this gap exposes researchers to acute physical danger, extreme psychological strain, and high logistical cost, while still failing to produce the temporal continuity the science demands.

1.2 Objectives

The primary objective of this project is to design and demonstrate a low-cost, autonomous bipartite marine environmental monitoring system that can be deployed from *vessels of opportunity* — including fishing vessels, sailing yachts, and research cruises — by non-specialist operators, thereby removing the need for dedicated human presence in hazardous Antarctic conditions. The system must:

1. Acquire continuous optical and physical sensor data (chlorophyll-a fluorescence, temperature, and depth) as proxies for microbial community dynamics.
2. Store acquired data reliably in non-volatile memory under extreme sub-zero conditions for the duration of a dive.
3. Transfer the stored data wirelessly to a researcher’s laptop upon probe retrieval, without requiring

a physical port on the probe enclosure.

4. Process the retrieved data locally, using machine learning anomaly detection to identify ecologically significant events entirely offline.
5. Sustain operation for up to twelve months on a single primary battery set without human maintenance.

1.3 System Requirements

The system must operate across two extreme thermal regimes: seawater approaching -1.8°C during submersion and ambient air down to -30°C upon retrieval to the vessel deck. The probe enclosure must withstand shallow-water pressure without water ingress, and all electronics must remain functional throughout. The complete system must fit within a 50 mm outer-diameter PVC pipe housing, be hand-deployable by a single operator, and have a total component cost not exceeding R2000. Data transfer between the probe and the laptop must complete within 60 seconds of connection, and the Mission Control graphical user interface must operate fully offline without any dependency on internet connectivity.

1.4 Scope and Limitations

This project addresses the sensor-to-laptop data pipeline and does not extend to long-range satellite transmission to mainland research facilities, which is identified as future work. The prototype targets near-surface deployments of up to 2 m depth; the design is architecturally scalable to greater depths but hardware validation at depth is beyond the present scope. Dissolved oxygen and nutrient sensors are excluded due to budget constraints, with chlorophyll-a fluorescence retained as the primary microbial proxy. Wireless charging of the probe battery, initially proposed as part of the docking station concept, was descope following the removal of the docking station from the architecture; the probe instead uses a primary Lithium Thionyl Chloride battery rated for twelve months of autonomous operation.

1.5 Report Outline

This report is structured as follows. Chapter ?? presents the literature review, surveying candidate data transmission protocols, embedded memory architectures, extreme-cold power systems, and marine sensing and housing methodologies to ground each design decision in the existing body of knowledge. Chapters ?? through ?? cover the four engineering subsystems in depth: Data Transfer and Visualisation (Chapter ??), Edge Architecture and Explainable AI (Chapter ??), Power and Thermal Management (Chapter ??), and Sensors and Housing (Chapter ??). Each chapter follows a consistent structure: requirements analysis, design choices with justification, final implementation, and acceptance test results. Chapter 7 draws overall conclusions and Chapter 8 identifies directions for future work.

Chapter 2

Literature Review

2.1 Introduction and Operational Context

The Antarctic marine ecosystem plays a critical role in global climate regulation, with marine microbes acting as the ocean’s primary early warning system for environmental shifts and pollution. However, gathering continuous, real-time biological data in this region is severely constrained by the ‘winter gap’, a period where extreme cold, sea-ice expansion, and severe weather make traditional research cruises prohibitively dangerous and expensive. Consequently, the reliance on infrequent manual sampling exposes marine researchers to immense physical and psychological risks while leaving vast spatial and temporal gaps in the ecological data pipeline.

To address this observational bottleneck without relying on highly complex, fully autonomous ocean gliders, a bipartite measurement system deployed from ‘vessels of opportunity’ (such as research cruises, fishing vessels or sailing yachts) presents a highly feasible engineering solution. This concept comprises a deck-bound docking station and a submersible measurement probe. The probe is lowered into the ocean via a tether to log localised environmental data, after which it is retracted to the docking station to securely transfer the telemetry for advanced on-board processing.

Operating a sophisticated electronic system in an environment where ambient temperatures drop below -30°C and seawater approaches -1.8°C introduces severe, multi-disciplinary engineering challenges. This literature review evaluates current academic research to inform the design of this bipartite system. The review is structured to analyse four critical domains: short-range data transmission protocols to bridge the two subsystems (Section 1.2), edge computing architectures and robust embedded memory to manage the massive datasets (Section 1.3), specialised battery chemistries and thermal management techniques to ensure survivability (Section 1.4), and the physical sensing modalities, waterproofing, and tethering mechanics required for reliable marine deployment (Section 1.5).

2.2 Data Transmission Subsystem

Operation of electrical systems in the extreme environmental conditions of Antarctica presents numerous engineering challenges. Liu et al. identifies sensor drift, packaging cost, and sea-water corrosion as significant factors that affect system performance in such climates [1]. Similarly, Uhm et al. highlights the necessity of active temperature control and thorough insulation of internal hardware to sustain the prolonged operation of a robotic system tested in Antarctica [2]. These findings highlight the dangers

posed by exposed ports or incomplete enclosures, which can allow the harsh Antarctic environment to compromise internal components. The following subsections examine candidate data transmission methods for communication between the measurement probe and the main docking station subsystem.

2.2.1 Wired Protocols

A tethered connection between the measurement probe and the docking station was initially proposed to enable real-time data communication. However, several critical concerns were identified: the potentially large and variable distance a tether would need to span depending on whether the solution is deployed on a large research vessel or a smaller fishing vessel; signal integrity degradation over long cable runs; and the exorbitant cost of thermally insulated, electrical cabling needed to transmit data in the Antarctic cold. Additional concerns included the risk of undetected faults developing over time due to limited maintenance access, which could result in silent data loss or corruption. For these reasons, long-range tethered solutions were ruled out in favour of short-range wired protocols to transfer data between the two subsystems once the measurement probe has returned from deployment. Table 2.1 summarises the characteristics of several candidate short-range wired protocols [3].

Name	Duplex	Max speed (Kbps)	Max distance (ft)
RS-232	Full	20	30
RS-422	Half	10000	4000
RS-485	Half	10000	4000
I^2C	Half	3400	<>
SPI	Full	> 1000	<>

Table 2.1: Short-range wired protocol comparison [3]

A further short-range protocol not listed in Table 2.1 is [Universal Serial Bus Type-C \(USB-C\)](#), which has become a widely adopted industry standard owing to its plug-and-play capability, high data transfer speeds, and versatility across connection types [3].

2.2.2 Wireless Protocols

Because the measurement probe operates while submerged and the docking station remains above the waterline, long-range wireless communication between them during deployment is not feasible. This is primarily due to the pronounced difference in electromagnetic signal propagation between water and air, which severely attenuates radio-frequency signals at depth [4]. Consequently, a short-range wireless protocol - employed once the measurement probe has surfaced and returned to the docking station - represents the most viable wireless approach. A long-range wireless solution was also considered, following the approach of Porte et al., who employed [Near Vertical Incidence Skywave \(NVIS\)](#) to relay data directly to a remote research station [5]. This architecture would have eliminated the need for a docking station subsystem entirely by transmitting data directly to a research base. However, it would have imposed significantly greater power, environmental protection and hardware requirements on the measurement probe, including the need for the either human personnel or another subsystem to ensure the positioning for [NVIS](#) to take place [5]. Although this approach was not adopted for

the measurement probe, it remains a viable option for the docking station subsystem to relay data onward after the initial sensor-to-docking-station transfer. Table 2.2 presents a comparison of candidate short-range wireless protocols, based on the comparative study conducted by Vidakis et al. [6].

Name	Range (m)	Frequency	Transmission time (s)	Power consumption
Bluetooth	100	2.4 GHz	24 Mbit/s	Medium
BLE	60	2.4 GHz	2 Mbit/s	Low
NFC	< 0.2	13.56 MHz	424 kbit/s	Low
Wi-Fi Direct	100	2.4 GHz, 5 GHz	250 Mbit/s	High

Table 2.2: Short-range wireless protocol comparison [6]

Of the protocols considered, [Near-Field Communication \(NFC\)](#) best satisfies the data transfer requirements of this application while minimising power consumption. By utilising a pair of inductive coils, an [NFC](#) interface can serve a dual function: transferring the recorded sensor data and simultaneously providing wireless charging capability [7]. Critically, this eliminates the need for a physical charging and data port on the measurement probe, which in turn permits a fully sealed enclosure and enhanced protection against the Antarctic environment. Although [Simultaneous Wireless Power and Data Transfer \(SWPDT\)](#) - the concurrent transfer of power and data over a single inductive link - remains an active area of research and is considered beyond the scope of this project [8], sequential operation (data transfer followed by charging, or vice versa) is readily achievable with existing [NFC](#) technology.

2.3 Edge Architecture and Data Management

The acquisition of high-fidelity environmental data by the submersible probe represents only the first phase of the system’s operational pipeline. Once gathered, this data must be securely stored, efficiently managed, and rapidly analysed to provide actionable ecological insights without overwhelming limited communication networks. This section investigates the hardware and software architectures required to process the massive datasets generated during Antarctic deployments. First, it explores robust embedded memory solutions capable of preserving data integrity in extreme sub-zero conditions before the probe docks. It then examines the necessity of edge computing frameworks, which bypass severe satellite bandwidth limitations by migrating computational tasks to the surface-level docking station. Finally, it reviews the application of machine learning and deep learning algorithms for in situ data processing, which enables the autonomous detection of microbial community shifts and environmental anomalies directly on the vessel.

2.3.1 Embedded Memory Solutions

The Antarctic marine monitoring system relies on a submersible measurement device that must continuously record and preserve critical biological and oceanographic data (such as conductivity, temperature, depth, and dissolved oxygen) before ascending to the surface docking station. Operating in seawater temperatures that can drop to -1.8°C and experiencing atmospheric exposure down to -30°C upon retrieval, the embedded memory must guarantee absolute data integrity. Traditional volatile memory is unsuitable due to data loss upon power failure, which necessitates the use of non-volatile

storage. However, conventional commercial storage solutions, such as standard SD cards based on NAND flash, are highly susceptible to thermal degradation, reduced write endurance, and charge leakage in extreme environments.

To achieve maximum system reliability, compactness, and low power consumption, [Embedded Non-Volatile Memory \(eNVM\)](#) directly integrated into the microcontroller die is highly advantageous. A comprehensive architectural review of [eNVM](#) highlights the trade-offs between legacy technologies, such as EEPROM and Flash [9]. While floating-gate EEPROM allows for byte-level rewriting, ideal for precise sensor data logging, it suffers from slower write speeds and higher power consumption compared to Flash [9]. Conversely, NAND flash offers higher density but operates using charge pumps to trap electrons on an electrically isolated gate. In extreme cold, charge mobility decreases, which can severely impede flash read/write operations and increase processing latency.

To mitigate the thermal vulnerabilities of charge-based storage, emerging [eNVM](#) technologies such as [Magnetoresistive Random Access Memory \(MRAM\)](#) present a superior alternative. [MRAM](#) stores data through magnetic states utilising the Tunnelling Magnetoresistance (TMR) effect rather than trapped electrons. Recent evaluations of embedded memories in cryogenic and extreme sub-zero temperatures (down to 77K) demonstrate that [Spin-Transfer Torque MRAM \(STT-MRAM\)](#) yields significantly higher read-voltage sensing margins in freezing conditions compared to standard SRAM [10]. By relying on magnetic polarisation, [MRAM](#) achieves near-infinite read/write endurance and maintains reliable data retention even below -40°C [10], which makes it a highly robust candidate for the Antarctic probe.

For the most extreme operational limits, purely mechanical data storage provides an innovative fail-safe. Joshi et al. detail the fabrication of a non-volatile [Micro-Electro-Mechanical Systems \(MEMS\)](#) switch designed specifically for harsh environments [11]. Encapsulated within a standard CMOS back-end-of-line process, this memory cell utilises a submicron refractory metal cantilever with two mechanically stable states. Data is stored physically through van der Waals stiction rather than electrical charge [11]. Because it lacks a charge-storage mechanism, this [MEMS](#)-based memory is fundamentally immune to temperature-induced charge leakage and operates reliably across an extreme thermal spectrum of -150°C to 350°C [11].

Ultimately, for the tethered Antarctic measurement device, relying solely on commercial flash memory introduces an unacceptable risk of data corruption during prolonged sub-zero deployments. The literature indicates that transitioning the probe’s on-board storage to [MRAM](#) architectures [10], or integrating [MEMS](#)-based non-volatile memory cells [11], provides the necessary combination of zero-power data retention, extreme thermal resilience, and high write endurance. This ensures that the captured microbial datasets securely survive the harsh conditions and are fully intact for inductive [NFC](#) transfer upon mating with the deck-bound docking station.

2.3.2 Edge Computing on Marine Hubs

Once the submersible probe safely records biological data to its non-volatile memory and returns to the surface, the system faces a severe communication bottleneck. Marine research in the Antarctic region relies almost exclusively on satellite communications to transmit data back to mainland laboratories. However, transmitting massive gigabyte-scale datasets, such as high-resolution optical fluorescence logs or continuous conductivity, temperature, and depth (CTD) profiles, over these networks is prohibitively

expensive, slow, and highly susceptible to latency and intermittent connectivity. As highlighted in a comprehensive review of oceanographic data processing [12], classical cloud-centric architectures fundamentally fail in remote marine environments due to these bandwidth limitations. To ensure real-time environmental insights, computational power must be relocated closer to the data source, which is an approach known as edge computing.

Implementing an edge-centric architecture perfectly aligns with the proposed vessel of opportunity concept. Recent research on marine [Internet of Things \(IoT\)](#) network topologies proposes a three-layer edge architecture tailored specifically for underwater and surface environments [13]. In this paradigm, the deck-bound docking station acts as the critical intermediate ‘edge node’ situated between the perception layer (the submersible probe) and the cloud layer (mainland servers). By offloading the raw datasets from the probe via [NFC](#) and processing them locally on the vessel, the docking station drastically reduces the payload size. Only critical ecological anomalies, summaries, or processed metadata are transmitted over the satellite link, which fundamentally bypasses the bandwidth bottleneck [12, 13].

Beyond simply filtering data, the edge hub must manage the overall data lifecycle and system infrastructure. Chave et al. detail the necessity of a robust [Data Management Subsystem \(DMS\)](#) for integrated ocean observatories [14]. A highly functional [DMS](#) does not merely pass data along, but it also handles local archiving, bidirectional data routing, and metadata tagging [14]. In the context of the Antarctic deployment, the docking station’s edge architecture will append crucial contextual metadata, such as real-time GPS coordinates from the host vessel, weather conditions, and time-stamps to the biological data retrieved from the probe, which ensures absolute data provenance before long-term storage or transmission.

Furthermore, edge computing resolves a critical energy conflict within the system. Processing complex biological datasets requires significant computational overhead that would rapidly drain the submersible probe’s battery. To maximise the probe’s endurance, the system can utilise task collaborative migration methodologies. Research into [Edge Computing-RESTful \(ECR\)](#) architectures for marine [IoT](#) demonstrates how heavy processing tasks can be dynamically scheduled and migrated from resource-constrained end-devices to more powerful, surface-level edge gateways [15]. By centralising the intensive data processing algorithms on the docking station, which has access to the vessel’s more stable power supply, the submersible probe can dedicate its limited energy budget entirely to maintaining sensor fidelity and executing the [NFC](#) data transfer.

Ultimately, designing the docking station as an intelligent marine edge hub rather than a passive charging cradle ensures that the massive datasets generated during the Antarctic winter gap are aggressively filtered, locally archived, and efficiently transmitted, all while preserving the power constraints of the remote sensing hardware.

2.3.3 Advanced Processing for Microbial Data

Once the submersible probe’s data is successfully transferred to the docking station via [NFC](#), the edge architecture must rapidly process the raw telemetry to extract meaningful ecological insights. A core objective of the Antarctic deployment is to serve as an early warning system for microbial community shifts and harmful environmental events. Traditionally, identifying these shifts required physical

water samples to be manually analysed in a laboratory setting. However, recent advancements have demonstrated that [Machine Learning \(ML\)](#) algorithms can successfully interpret complex, non-linear relationships in chemical and biological oceanographic data directly in situ [16].

Because the proposed measurement device utilises optical and fluorescent microsensors to detect dissolved organic matter, the data output consists of raw optical radiance and impedance values rather than explicit biological classifications. A review of [ML](#) applications in ocean colour and optical sensing highlights the efficacy of [Artificial Neural Network \(ANN\)](#)s and [Support Vector Machine \(SVM\)](#)s in translating this raw optical backscatter and fluorescence into accurate biological parameter estimates, such as phytoplankton concentration and microbial biomass [17]. By hosting these trained regression models on the docking station, the system can autonomously convert physical sensor voltages into quantitative biological data.

However, marine ecological datasets gathered in extreme environments are notoriously noisy. Sensor data can be easily skewed by ice particulates, biofouling on the optical lenses, or rapid localised temperature fluctuations. To address this, researchers are increasingly unlocking the potential of [Deep Learning \(DL\)](#) for marine ecology [18]. Off-the-shelf [DL](#) architectures, such as [Convolutional Neural Network \(CNN\)](#)s applied to one-dimensional time-series data, have proven highly effective at filtering out environmental noise and recognising cryptic patterns in ecological datasets that traditional statistical methods overlook [18].

The ultimate operational goal of the docking station’s software is anomaly detection, which is identifying the moment a baseline microbial community undergoes a stress-induced shift. Designing algorithms for this specific task is uniquely challenging. As outlined in recent studies on building [ML](#) challenges for scientific anomaly detection, models must be carefully trained to distinguish between a true scientific outlier (genuine ecological event) and a spurious anomaly (sensor malfunction or background noise) [19]. In the context of the Antarctic winter gap, an unsupervised machine learning model, such as an Isolation Forest or an Autoencoder, can be deployed on the edge hub to continuously monitor the incoming data.

By defining the ‘normal’ winter microbial baseline, the edge algorithm can flag significant deviations in real-time [19]. Consequently, the docking station does not merely archive data, but it also actively diagnoses the ecosystem, which ensures that only high-value, critical alerts regarding microbial community shifts are transmitted over the constrained satellite networks back to mainland researchers.

2.4 Extreme Cold Power System

In Antarctica, the environmental conditions present challenges that engineers need to consider when designing measurement devices. Some of these challenges include the ocean temperature. The Antarctic waters fall near temperatures of -1.8 degrees Celsius, and ambient temperatures falling below -30 degrees Celcius quite frequently. These conditions, specifically, degrade thermal stability, storage of energy, as well as the power delivery in engineering systems. For the two-part system consisting of both a sensing submersible probe and a deck-mounted docking station, the reliable delivery of power to both systems becomes the primary constraint of long-term operation. Conventional systems that could be used in this environment would experience difficulties with increased internal resistance, a

reduction in capacity, as well as thermal inefficiencies relating to the operation of the device. Therefore, advancements in battery chemistry, thermal management, and efficient energy system design should be considered.

2.4.1 Battery Chemistry at Sub-Zero Temperatures

Batteries such as lithium-ion and sodium-ion will be considered. Both these batteries experience quite a significant amount of performance degradation at low temperatures (such as the temperatures in Antarctica). This performance degradation is mainly due to reduced ionic conductivity and an increased amount of charge transfer resistance. Specifically, electrolytes have an important role within the battery chemistry at low temperatures, as their ion mobility, viscosity, and melting point directly affect the battery efficiency [20]. Standard carbonate electrolyte batteries have performance issues at low temperatures, which include reduced conductivity and a large increase in interfacial resistance [20]. Current research into the field of alternative electrolyte systems suggests that carboxylate ester-based electrolytes are highly effective at low temperatures, as they possess low viscosity, low melting points (below -70 degrees Celcius), and high dielectric constants [20]. This makes them highly favourable for sub-zero environments such as Antarctica. The properties mentioned above increase the ion transport and reduce the overall charge transfer resistance, hence, improving the performance of the battery in extreme cold temperatures [20]. However, there are challenges presented when stabilizing electrode-electrolyte interfaces due to the increase in reactivity at low temperatures [20]. In addition to this electrolyte enhancement, the battery thermal sensitivity should also be considered. Batteries such as the lithium-ion cell are extremely dependent on temperature. Extreme cold can lead to reduced capacity as possible safety risks such as fire and explosion [21]. Therefore, there is a need for integrated thermal regulation strategies.

2.4.2 Thermal Management in Embedded Systems

Thermal management of battery systems is integral in controlling the battery efficiency, preventing thermal runaway (battery heating faster than it can dissipate the heat) [21], and ensuring the battery can operate reliably at low temperatures. Lithium-ion batteries in particular suffer from cold temperatures, which lead to poor heat dissipation, capacity fade, non-uniform cell behaviour, and possible catastrophic battery failure [21]. According to W. Wu et al., an efficient and effective passive thermal management system using **Phase Change Materials (PCM)**. The **PCM** absorb and release latent heat during the change of phases [21]. This maintains relatively stable operating temperatures. Furthermore, W. Wu et al. suggest that composite **PCM** systems improved with materials such as graphite significantly improve the thermal conductivity as well as the heat dissipation [21]. This improves the stability of temperature across battery modules [21]. In addition to the internal temperature control of the battery, external insulation also plays a crucial role in battery survival. This would minimize heat loss in cold environments in the Antarctic, such as ship decks. According to Shahar Idan and Yuri Feldman, further researched insulation strategies, such as porous and structured materials, significantly reduces convective heat transfer by suppressing the internal air circulation and limiting temperature difference [22]. Furthermore, M. Rajendiran et al. suggest that additional insulation materials, such as nitrile rubber foam, show effective thermal resistance as well as increased thermal retention compared to vacuum insulation in some configurations, specifically when the material's heat capacity and surface

temperature effects are taken into account [23]. Thermal stability is essential for sensing processes such as [Loop-Mediated Isothermal Amplification \(LAMP\)](#), which require stable operating temperatures, hence the need for hybrid insulation solutions [24].

2.4.3 Power Budgeting and Duty-Cycling

Power management within battery operation is essential for extending the battery lifetime within energy-constrained Antarctic conditions. Wireless sensor nodes, as well as embedded systems, usually operate in low-duty-cycle modes. Most of the time, these systems are spent in sleep states, which draw power. Even during sleep states, particularly when looking at voltage regulation components, can significantly reduce battery life [25]. According to D. Balsamo et al., adaptive duty-cycling techniques for DC-DC converters can provide an effective solution to this problem. This can provide dynamic control of converter activation based on the demand of the system as well as the system leakage. The process of intermittently activating the system using internal capacitor storage and only activating the converter when needed will significantly reduce the quiescent loss and hence extend battery life [25]. This approach is particularly beneficial in systems such as remote sensing nodes where sleep-mode battery consumption dominates the overall energy usage. Furthermore, the reduction of energy consumption during communication between nodes is essential. Solutions such as low-power communication protocols and improved scheduling will further reduce the overall energy demand. Communication protocols also play a critical role in overall system energy consumption. Standard wireless technologies such as [Wireless Fidelity \(Wi-Fi\)](#), Bluetooth, and ZigBee all consume a high amount of power when in use and have limited coverage [26]. Contrastingly, technologies known as [Low-Power Wide-Area Network \(LPWAN\)](#), which include [Long Range Wide Area Network \(LoRaWAN\)](#), enable long-range communications with significantly reduced power consumption. Experimental implementations have shown that [LoRaWAN](#) systems can reduce power consumption by 18-20 times compared to [Wi-Fi](#), while also providing encrypted communications [26]. The properties mentioned above make [LoRaWAN](#) a well-suited and low-power solution needed for Antarctic devices. These solutions provide low-power communication and adaptive duty-cycling, hence enabling the system to achieve long-term operation in Antarctica.

2.5 Marine Sensing Modalities and Deployment Mechanics

The physical implementation of the submersible probe requires careful consideration of three interconnected systems. The sensors that acquire environmental and microbial data, the mechanical systems that enable deployment from a moving vessel, and the waterproof housing that protects sensitive electronics in the harsh Antarctic marine environment. This section reviews the literature across these domains to inform the design of a compact, low-cost, and user-friendly probe suitable for deployment from vessels of opportunity.

2.5.1 Microbial Detection Sensors

The selection of sensors for the submersible probe must balance scientific relevance, power consumption, cost, and ease of use. Emma Rocke’s primary interests are oxygen, temperature, salinity, and nutrients, with the ‘added bonus’ of capturing microbial information where possible. The literature reveals two

primary approaches for in-situ microbial sensing: optical methods that measure proxies for microbial biomass, and electrochemical methods that detect microbial activity or specific compounds. Optical sensing technologies offer the most mature and field-ready approach. Raimbault and Sanchez Olvera [27] demonstrated the Openprobe, a low budget probe that integrates conductivity, temperature, depth ([Conductivity, Temperature, Depth \(CTD\)](#)), dissolved oxygen, turbidity, and chlorophyll-a fluorescence. The chlorophyll-a fluorometer uses synchronous detection to reject ambient light which is critical for near surface readings where sunlight penetration is significant and captures the weak fluorescence signals from in-vivo chlorophyll. The oxygen sensor employs an optode based on phase-shift measurement, similar to industry-standard Aanderaa sensors, which provides robust performance without the oxygen consumption issues that plague Clark-type electrodes. Critically, Kühl [28] notes that such oxygen optodes are solid-state sensors that can handle sub-zero temperatures and have been successfully deployed in sea ice environments, directly addressing the Antarctic operational constraint.

The practical path forward, therefore, is to adopt the optical sensing approach demonstrated by the Openprobe. This provides Emma with the core environmental parameters she requires, plus chlorophyll-a as a proxy for phytoplankton biomass which provides valuable insight into microbial community dynamics without the complexity of whole-cell biosensors. The Openprobe’s use of an Adafruit Feather microcontroller and custom [Printed Circuit Board \(PCB\)](#)s demonstrates that such a system can be built with accessible, low-cost components.

2.5.2 Tethering and Winch Mechanisms

The deployment of a submersible probe from a moving vessel of opportunity requires careful attention to tether management. Peraud et al. [29] provide a comprehensive analysis of this challenge in their development of a smart tether winch for an [Remotely Operated Vehicle \(ROV\)/Unmanned Surface Vehicle \(USV\)](#) system. They identify two critical problems with tether systems, namely drag forces that increase with vehicle motion and water current, and entanglement risks that can compromise safety. Their solution is a two-part system consisting of a sensor near the [ROV](#) and an intelligent winch on the surface vessel which maintains a small amount of tension in the tether that minimizes both issues. Peraud’s tension module uses toothed wheels to grip the tether with precise speed control via optical encoder feedback, while a separate torque-controlled spooler ensures tidy cable storage. The use of microswitches to detect tether departure angle offers a low-cost method for understanding probe orientation relative to the vessel. Peraud’s work shows the need for tether management while simultaneously highlighting the complexity of achieving it. Their system requires two motors, multiple sensors, and a sophisticated ROS2-based control architecture.

The smart winch literature demonstrates that active tether control is a complex problem requiring significant hardware and software investment. For our system, the tether serves only as a physical connection, with wireless data transfer and recharging occurring upon docking. This allows us to prioritize simplicity implementing a semi-automated winch with the aim of reducing complexity and cost while maintaining safety. This aligns with the vessels of opportunity operational constraints identified by the Ship Observations Team [30] which advise that equipment must be ‘as unobtrusive as possible,’ require minimal ship modifications, and be operable by non-specialists. A straightforward tether management approach ensures that vessel operators can deploy and recover the probe with minimal training, without the burden of maintaining a complex smart winch system.

2.5.3 Waterproofing and Pressure Housings

The protection of electronics in the marine environment, particularly in Antarctic conditions, requires careful material selection and sealing design. The literature provides several validated approaches across different cost and complexity levels. For shallow-water deployments, PVC and 3D-printed housings offer a low-cost entry point. The Openprobe project [27] used a Blue Robotics 2-inch diameter enclosure tube which is rated to significant depths. They also demonstrated the use of [Stereolithography \(SLA\)](#) 3D printing for custom end caps that integrate O-ring grooves, optical ports, and sensor alignment features. The use of stereolithography (SLA) over fused deposition modeling (FDM) is critical because SLA produces fully dense, watertight parts with the dimensional accuracy required for O-ring sealing.

For optical sensors, which form the core of the proposed sensing suite, the waterproofing challenge extends to the optical ports. The Openprobe uses polydimethylsiloxane [Polydimethylsiloxane \(PDMS\)](#), a silicone-based material with good transparency in the visible and near-infrared spectrum, to create optical windows and as a potting compound. The SCINI [ROV](#), developed for under-ice deployment in Antarctica [31], uses an even simpler approach where [Light Emitting Diode \(LED\)](#)s and lasers are cast directly in epoxy resin, creating a solid, waterproof optical assembly without the need for separate pressure housings. This technique has been field-validated to depths of 330 meters in Antarctic waters and represents a robust, low-cost solution for our probe. For sealing, the literature consistently identifies O-rings as the standard method for underwater housings. Proper O-ring groove design, combined with appropriate material selection and hydrophobic grease provides reliable sealing when the housing is assembled correctly. The modular approach used by SCINI, with separate housings for cameras, electronics, and thrusters, simplifies maintenance: if one module fails, it can be replaced without disassembling the entire probe.

2.6 Conclusions from Literature Review

The literature demonstrates that successfully designing a marine monitoring system for the Antarctic winter gap requires a delicate balance between robust physical encapsulation, localised data processing, and highly resilient power management. To accommodate deployment from vessels of opportunity, the mechanical design must prioritise simplicity and usability. Rather than employing complex, actively controlled smart winches, a simplified semi-automated tethering mechanism allows non-specialist crews to deploy the submersible probe safely. At the sensor level, optical methodologies, specifically chlorophyll-a fluorometers and phase-shift oxygen optodes, provide the most field-ready approach to gathering critical biological data without the fragility of whole-cell biosensors. These sensors can be effectively protected using SLA 3D-printed housings, O-ring seals, and direct epoxy potting for optical windows.

To survive the extreme thermal operating constraints (-30°C to -1.8°C), the probe's power and memory systems must diverge from standard commercial architectures. Traditional carbonate-based lithium-ion batteries and NAND flash memory are highly susceptible to capacity fade and charge leakage in freezing conditions. The integration of carboxylate ester-based electrolytes, coupled with passive [PCM](#) thermal insulation, ensures reliable power delivery. Furthermore, the use of emerging non-volatile memory technologies, such as [MRAM](#) or [MEMS](#)-based switches, guarantees that the recorded environmental data remains uncorrupted during prolonged sub-zero deployments.

Once the probe returns to the surface, the deck-bound docking station serves as a critical edge-computing hub. Classical cloud-centric architectures fail in remote marine environments due to the extreme bandwidth limitations and high latency of satellite communications. By offloading complex processing tasks to the docking station, the system utilises Machine Learning (ML) and Deep Learning (DL) algorithms, such as Convolutional Neural Networks and Isolation Forests, to autonomously filter noisy optical data and detect true ecological anomalies. Consequently, only high-value microbial community shifts are transmitted via satellite, which drastically reduces payload size and preserves system energy.

To bridge the submersible probe and the edge-computing docking station without compromising the waterproof enclosure, NFC has been selected as the short-range data transmission medium for this subsystem. By utilising inductive coupling, NFC offers both data offloading and wireless charging in a fully encapsulated portless form factor. This selection directly addresses the environmental robustness requirements identified in the literature, which eliminates the corrosion and freezing risks associated with exposed electrical contacts. If NFC proves insufficient to meet the system’s data throughput or range requirements under operational conditions, USB-C will serve as a fallback solution, again providing a combined data transfer and charging interface. Ultimately, this synthesised approach yields a secure, power-efficient, and highly autonomous observational pipeline capable of transforming how marine microbiological data is gathered in the world’s harshest environments.

Chapter 3

Data Transfer and Visualisation

[Author: Mogamad Saeed Solomon - SLMMOG032]

3.1 Introduction

This subsystem deals with the transfer of the stored data on the Antarctic probe to a Local Mission Control [Graphical User Interface \(GUI\)](#) running on a laptop for the data to then be visualized before being sent to the [ML](#) processes described in the following chapter. The aim of this subsystem is to provide an accurate, reliable and low-power data communication channel between the probe and laptop, and to implement a [User interface \(UI\)](#) for the researchers to interact with the retrieved data. The subsystem comprises two software components, `Backend.py` and `GUI.py`, alongside firmware running on two ESP32-C6 microcontrollers: the submersible probe and a receiver dongle that interfaces with the laptop over USB serial.

3.2 Requirement Analysis

When the client, *Emma Rocke*, was interviewed one of the main points that arose was the need to collect raw data samples to do post-processing at a later stage. To meet this requirement, large volumes of raw data need to be accurately and reliably transferred between the two parts of the system while adhering to the battery limitations of prolonging the system lifespan. Due to the extreme environmental conditions of the Antarctic, a fully encapsulated solution is required, alluding to the need for wireless communication between the two parts of the system. Aside from purely visualizing the data, because researchers or civilians will be operating the system it needs to be intuitive and easy to interact with, control, and monitor. As such, the requirements for this subsystem are listed below in [Tables 3.1](#) and [3.2](#).

3.2.1 Functional Requirements

Table 3.1: System Functional Requirements

ID	Requirement	Description
FR1	Quality	The data transfer needs to be accurate, fast and reliable
FR2	Wireless transfer	Data transfer from probe to laptop needs to be wireless
FR3	Error checking	No corruption of data during transfer needs to be implemented
FR4	Save/Load data	Functionality to save current data or load previous data
FR5	Wake-up	The probe needs to wake up from deep sleep to enable its antennae for advertising and vice versa
FR6	Offline	The transfer protocol and GUI needs to be able to function without any internet connection

3.2.2 User Requirements

Table 3.2: System User Requirements

ID	Requirement	Description
UR1	GUI	Interactive GUI for users to visualize the retrieved data on the receiver end (Laptop/Host).
UR2	Probe setup	Easy, intuitive control interface for the probe to control its mode of operation (e.g., Prepare Dive vs. Data Offload).
UR3	Battery indicator	A means to determine the remaining probe battery life to prevent scenarios where the probe could potentially fail mid-exploration.
UR4	Data display	Display raw data retrieved from the exploration in both a raw format and a more intuitive way (such as via graphs or charts).

3.2.3 Specification Analysis

Table 3.3: Specification analysis

ID	Specification	Description
SP01	Wake-advertising time	The submersible probe needs to wake from deep sleep to advertise its signal in $< 10s$
SP02	Data transfer window	The complete data transfer process needs to be $< 1min$
SP03	Data integrity	Transferred data accuracy $> 95\%$
SP04	GUI record update time	The time taken to update the display upon receipt of a record needs to be $< 1s$ to prevent lag
SP05	Battery indicator	A battery level status indicator needs to be visible on the GUI for clients to be aware of the probe remaining battery lifespan

3.2.4 Traceability Matrix

Table 4.4 maps each requirement to the design decisions and ATPs that address it.

Table 3.4: Traceability Matrix

Requirement	Specification	Acceptance Test Procedure (ATP)
FR1	SP04	ATP06
FR2	SP02	ATP01
FR3	SP03	ATP02
FR4		ATP04
FR5	SP01	ATP03
FR6	SP04	ATP05
UR1	SP04	ATP04, ATP06
UR2		
UR3	SP05	ATP07
UR4	SP02, SP04	ATP04

3.3 Acceptance Test Procedures

The ATPs section outlines the tests to be conducted on the final design alongside their acceptance criteria to determine whether the specifications based on the requirements have been met.

Table 3.5: ATP for the Data Transfer and Visualisation subsystem

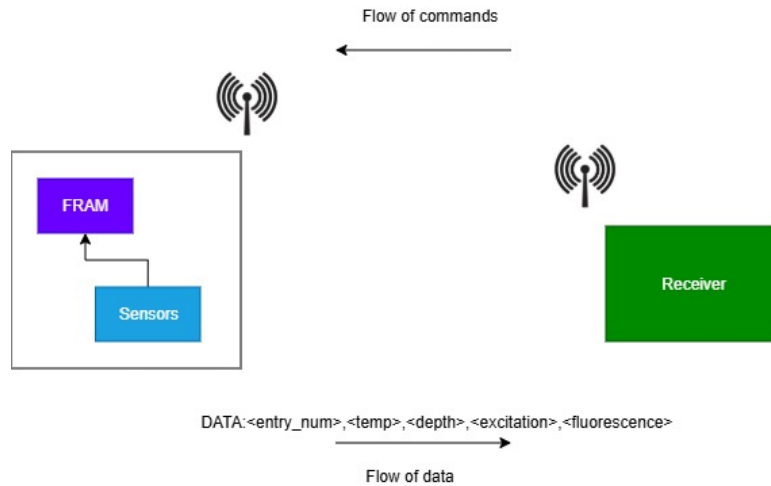
ID	Specification Description	Acceptance Criteria
ATP01	Wireless recovery	Transfer distance between sender and receiver at least 10 m
ATP02	Data integrity > 95%	Data records accurately transferred with a final error margin < 5%
ATP03	Initiate transfer on wake-up in under 10 s	When the probe wakes from deep sleep the transfer protocol must be initialised in under 10 s and safely de-initialised before returning to sleep
ATP04	Data visualisation	Raw data received from probe/loaded able to be displayed on GUI
ATP05	Fully offline operation	Both the transfer protocol and GUI are able to be operated without any internet connection
ATP06	GUI response rate	GUI tables/graphs updated in less than a second for seamlessness
ATP07	Battery display	GUI indicates the remaining probe battery life

3.4 Design Choices

3.4.1 Block Diagram

Raw data is collected from the sensors of a submersible probe deployed overboard a vessel of opportunity and stored locally on the probe for the duration of the mission. Once retrieved, the probe initiates a wireless transfer process to send the raw data from its onboard storage to the receiver system, where post-processing and data viewing take place (FR2, UR1, UR4). A simplified overview is shown in Figure 3.1.

Figure 3.1: Simplified block diagram of flow of data throughout solution



3.4.2 Transfer Protocol Selection

Three wireless protocols were evaluated over the course of the project. The figures of merit used for comparison were range, power consumption, implementation cost, and suitability for use in the Antarctic marine environment.

Option 1: NFC

The initial design centred around a bipartite system comprising the submersible probe and a docking station for data offloading and charging. NFC was considered primarily because it allowed the probe to be entirely portless. The same 13.56 MHz coil could theoretically service both data transfer and inductive wireless charging, eliminating every penetration point in the enclosure and removing the primary ingress risk in a cold saltwater environment. At 424 kbit/s (NFC High Speed), the anticipated data volume of well under 1 MB per dive could be transferred in a few seconds.

However, this approach was ruled out for three reasons: integrated NFC wireless charging listener ICs (e.g. Renesas PTX30W [32]) were far too complex and expensive for the project budget; the docking station was descoped since vessels of opportunity may lack the means to power it; and the NFC protocol's strict proximity and orientation requirements are poorly suited to use on the open ocean.

Option 2: Bluetooth Low Energy (BLE)

BLE was the natural successor candidate. The ESP32-C6 includes a native BLE radio, incurring no additional hardware cost. BLE offered substantially higher data rates than NFC (up to 2 Mbit/s in BLE 5.0) and greater tolerance for device misalignment. A GATT server profile was fully designed and implemented on the probe, exposing four characteristics: **METADATA** (probe identity), **CONTROL** (operational state commands), **DATA** (transfer payload), and **STATUS** (real-time health monitoring). CRC-16 packet validation and GPIO-based reed switch deep sleep wakeup were also implemented at this stage.

Despite a complete firmware implementation, BLE failed during hardware validation. The probe’s BLE advertising was undetectable by both a smartphone running nRF Connect and a laptop running a Python GATT server, across both available ESP32-C6 development boards. Reversing the setup by making the laptop act as the server did not resolve the issue. Given that the problem manifested identically on both boards, the root cause was attributed to immaturity in the ESP32-C6 BLE stack, necessitating a more reliable alternative.

Option 3: ESP-NOW (Selected)

ESP-NOW is a connectionless protocol developed by Espressif that operates at the Data Link layer, reducing the five layer OSI model to one [33]. It is native to the ESP32-C6, requires a second ESP32 as a receiver dongle as its only additional cost, and proved immediately functional where BLE had failed. Table 3.6 summarises the full comparison across all three protocols.

Table 3.6: Wireless Protocol Comparison

Protocol	Range	Power	Data Rate	Cost	Notes
ESP-NOW	~100 m	Ultra-low	1 Mbit/s	R230 (2x ESP32)	Data-link layer operation; minimal stack overhead; native to ESP32-C6 [33]. Selected.
BLE	<10 m	Low	2 Mbit/s	R115 (ESP32)	Fully implemented; abandoned due to hardware-level advertising failure on both ESP32-C6 boards.
NFC	<0.2 m	Low	424 kbit/s	R508 (ESP32, NFC receiver and dynamic tag)[34, 35]	Strict orientation requirements unsuitable for marine use; docking station de-scoped.

ESP-NOW was selected on the basis of proven hardware reliability, the lowest effective implementation cost at R230, a range far exceeding the ATP01 requirement of 10 m, and ultra-low power consumption consistent with the probe’s battery constraints.

3.4.3 GUI Architecture

The Mission Control GUI was designed in Python using the CustomTkinter framework, with `matplotlib` embedded for graphing. Python was chosen for two reasons: the ML anomaly detection models in Subsystem 2 are also implemented in Python, enabling direct integration without an inter-process boundary; and PySerial provides straightforward cross-platform serial communication without driver dependencies beyond the USB CDC stack built into the ESP32-C6.

The GUI was architected as two decoupled modules: `Backend.py` handles all serial I/O, packet parsing, and ML evaluation, while `GUI.py` provides the interface layer and calls backend functions through a thread-helper pattern (`run_async` for async coroutines, `run_in_thread` for blocking I/O) to prevent the GUI from freezing during transfers. This separation made it possible to unit-test the backend independently using simulated data and allowed easier management of the growing codebase.

The initial GUI design is shown in Figure 3.2, highlighting the key elements that address the user requirements:

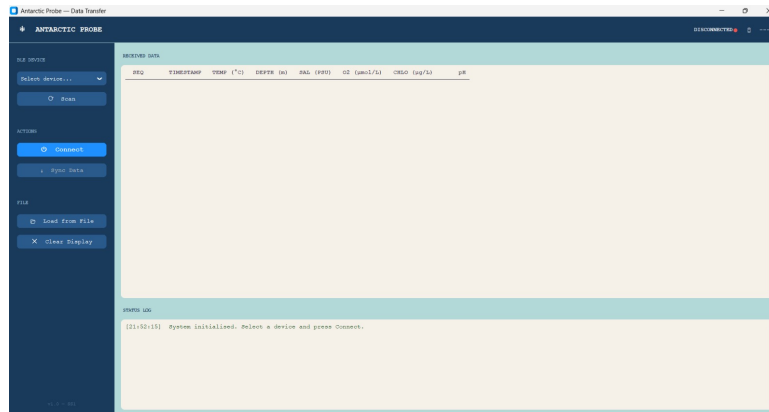


Figure 3.2: Initial GUI design

- Top panel indicating current probe status and battery level (UR3, SP04)
- Side toolbar for intuitive probe control (UR2)
- Display area for incoming raw data (UR4)
- Save/Load file functionality (FR4)
- Status terminal for task execution feedback (UR5)

3.5 Final Design

3.5.1 Probe Command-Controlled Structure

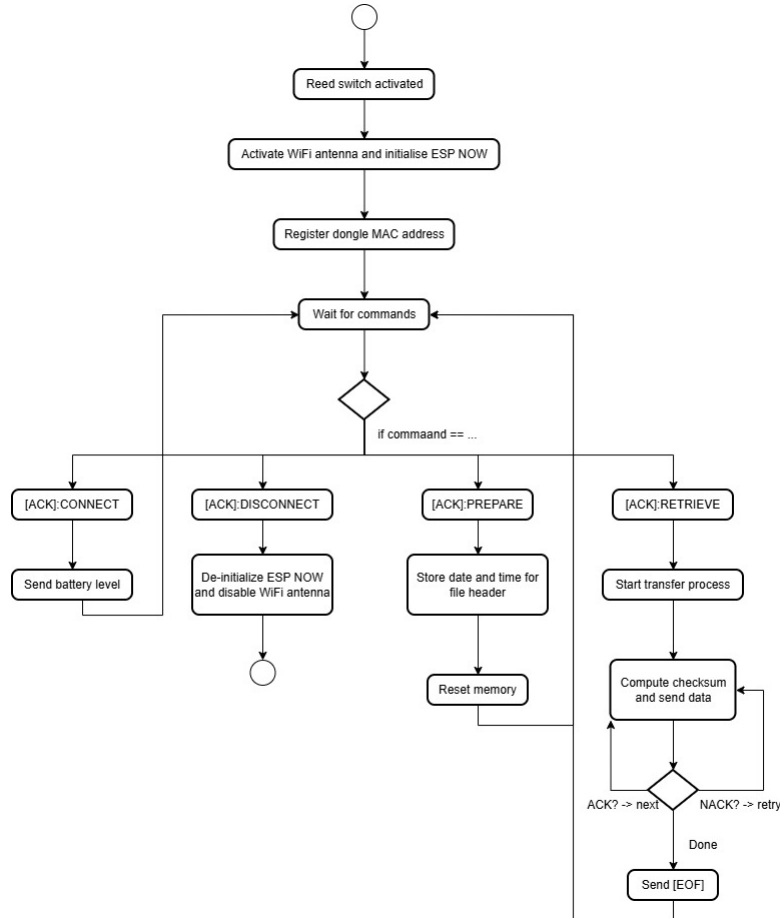
The probe wakes on a reed switch interrupt, initialises the WiFi radio in station mode, and calls `ESPNow_Init()` to register the dongle’s MAC address as a peer (FR5). It then waits for commands sent by the GUI via the dongle over ESP-NOW. The four supported commands are:

- **[CMD]:CONNECT** — Verifies communication between probe and dongle and updates the probe's remaining battery level on the GUI
- **[CMD]:DISCONNECT** — De-initialises ESP-NOW on the probe to conserve battery
- **[CMD]:PREPARE** — Resets the probe's memory and stores the current date and time in the session header, providing a timestamp reference for each record (since the probe cannot track time during deep sleep)
- **[CMD]:RETRIEVE** — Initiates the data offloading process for all records logged during the mission

The full command-controlled state flow is shown in Figure 3.3.

On the dongle side, `Receiver_Dongle.ino` implements a three-state machine (**IDLE**, **READY**, **CONNECTED**) controlled via a serial command protocol from the laptop. On boot, the dongle prints a heartbeat message allowing the Python backend to locate the correct COM port automatically. The backend then sends `CMD:STOP_HEARTBEAT`. When the connect button is pushed by the user a `CMD:CONNECT` is issued which initialises ESP-NOW and registers the probe as a peer. When `CMD:RETRIEVE` is issued, the dongle arms itself and forwards validated records to the laptop as `DATA: CSV` lines, consumed by `Backend.py`'s `sync_probe()` function.

Figure 3.3: Stateflow chart of the command-controlled flow of the submersible probe



3.5.2 Wire Format: `data_packet.h`

All packet definitions are centralised in `data_packet.h`, included by both the probe firmware (`espnow_transfer.cpp`) and the receiver dongle firmware. The shared header uses packed structs to ensure an identical binary layout on both ESP32 targets. Static assertions verify struct sizes at compile time (`SessionHeader_t`: 16 bytes; `ProbeRecord_t`: 62 bytes), catching layout regressions if field types are changed. This comfortably falls within the 250-byte maximum payload of ESP-NOW.

One record is transferred per message to keep error-checking straightforward and to allow additional sensor fields to be appended in future without redesigning the framing layer. The record structure is detailed in Table 3.7.

Table 3.7: Probe Record Packet Structure (62 bytes, little-endian)

Byte Offset	Type	Field	Description
[0]	uint8	<code>pkt_type</code>	Always <code>PKT_TYPE_RECORD</code> (0xA1)
[1:5]	uint32	<code>entry_num</code>	Record index (0-based)
[5:9]	uint32	<code>ms_since_start</code>	Milliseconds since <code>session_start_ms</code>
[9:13]	float32	<code>temperature_c</code>	Temperature (°C)
[13:17]	float32	<code>pressure_dbar</code>	Pressure (dbar)
[17:21]	float32	<code>spec1</code>	Spectrometer channel 1
[21:25]	float32	<code>spec2</code>	Spectrometer channel 2
[25:29]	float32	<code>spec3</code>	Spectrometer channel 3
[29:33]	float32	<code>spec4</code>	Spectrometer channel 4
[33:37]	float32	<code>spec5</code>	Spectrometer channel 5
[37:41]	float32	<code>spec6</code>	Spectrometer channel 6 (excitation)
[41:45]	float32	<code>spec7</code>	Spectrometer channel 7 (fluorescence)
[45:49]	float32	<code>spec8</code>	Spectrometer channel 8
[49:53]	float32	<code>spec9</code>	Spectrometer channel 9
[53:57]	float32	<code>spec10</code>	Spectrometer channel 10
[57:61]	float32	<code>spec11</code>	Spectrometer channel 11
[61]	uint8	<code>checksum</code>	XOR of bytes [0:61]

All multi-byte fields are little-endian, matching the native byte order of both the ESP32 (RISC-V based) and the x64 laptop. Chlorophyll-A concentration is deliberately excluded from the on-wire format: the probe transmits raw `excitation_raw` and `fluorescence_raw` ADC counts, and the conversion to concentration is performed by Subsystem 2’s [ML](#) processing step. This is to prioritise raw data collection and prevent power drain by performing computation on the probe.

The final byte is an XOR checksum computed over all preceding bytes (FR3), chosen as a lightweight integrity check that avoids per-record traffic overhead. A failed record is reattempted at most 3 times at 500 ms intervals; records that still fail are discarded and appear as empty entries in the raw data display.

3.5.3 Mission Control GUI

The GUI is a standalone Python desktop application (`GUI.py`) that runs without internet connectivity (FR6). It is structured around a top toolbar, a sidebar of controls, a tabbed main panel, and a persistent status log. The application state machine mirrors the dongle's: buttons are disabled until the dongle is detected, probe controls are then unlocked, and data-retrieval functions are enabled only once a probe connection is confirmed. The final design is shown in Figures 3.4 and 3.5.

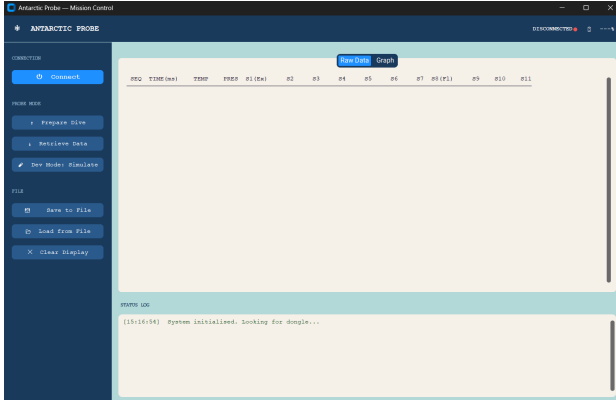


Figure 3.4: Final GUI design: Raw Data tab

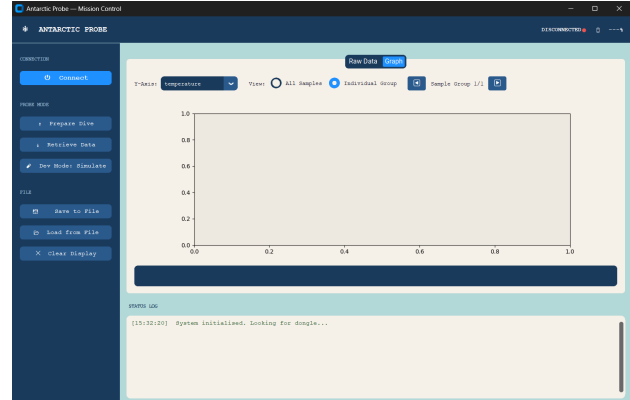


Figure 3.5: Final GUI design: Graph tab

Sidebar Controls

The sidebar is divided into three sections. The Connection section contains the Connect/Disconnect button, which triggers the dongle discovery routine and, on success, the probe pairing sequence. The Probe Mode section provides three action buttons: Prepare Dive, which sends **PREPARE**; Retrieve Data, which sends **RETRIEVE**; and Dev Mode: Simulate, which injects synthetically generated records directly into the GUI at the same rate and in the same format as a live retrieval, without requiring a physical probe or dongle. This served as the primary simulation submodule for validating the full GUI pipeline. This included live table updates, graph rendering, and anomaly flagging independently of hardware availability. The File section provides Save to File (CSV export), Load from File (CSV import), and Clear Display (FR4).

Raw Data Tab

The Raw Data tab displays records in a fixed-width monospaced table with columns for all fields in the transfer protocol. Records are appended live as they arrive; anomalous records are flagged inline with an [ANOMALY] marker produced by the `MarineAnomalyDetector` from Subsystem 2. The text widget is read-only to prevent accidental modification.

Graph Tab

The Graph tab provides a visual view of trends in the sensor data. A dropdown selects the y -axis variable (temperature, pressure, excitation, and so forth), with the x -axis fixed to elapsed time in milliseconds. Two view modes are available:

- **All Samples** — plots the complete dataset as a connected line series.

- **Individual Group** — divides the dataset into groups of 15 readings (configurable via `SAMPLES_PER_GROUP`) and allows the researcher to step through groups with previous/next buttons, enabling closer inspection of short time windows.

The graph updates automatically on y -axis or mode changes and refreshes incrementally as new records arrive during a live retrieval. Screenshots of the GUI displaying simulated data in both the raw data and graph views, including a dataset with injected anomalies, are provided in [Appendix A](#).

3.5.4 Backend: Serial Protocol and Data Flow

`Backend.py` manages the shared serial port and exposes a clean async/sync API to the GUI. On startup it scans all COM ports, reads up to five lines from each, and confirms the correct port by matching the `DONGLE_READY_MSG` string, making discovery automatic regardless of which USB port the dongle occupies. The `sync_probe()` function reads `DATA:` lines in a blocking loop, evaluates each record for anomalies via the `MarineAnomalyDetector`, and delivers results to the GUI's `on_record` callback in real time. The loop terminates on a `[SESSION] EOF` marker from the dongle.

3.5.5 Software Prerequisites and Execution

The GUI requires Python 3.10 or later. The following third-party libraries must be installed prior to first use. With an active internet connection, all dependencies can be installed in a single terminal command:

```
pip install pyserial customtkinter matplotlib
```

Once installed, the application is launched from a terminal in the directory containing `GUI.py`:

```
python GUI.py
```

No [Integrated Development Environment \(IDE\)](#) is required. After launch, no further internet access is needed and the application operates fully offline (FR6). The receiver dongle must be plugged in via USB before pressing Connect, after which the GUI will locate the correct COM port automatically.

3.6 Testing and Results

3.6.1 Testing Procedures

Due to the black-box testing mandate and the May 18th deadline, each component of the subsystem was tested independently before integration testing was attempted. Where hardware-in-the-loop testing was performed, each test was repeated across at least three independent runs and the results averaged to account for timing variability.

The end-to-end data transfer pipeline was verified by issuing a `CMD:RETRIEVE` command from the GUI to a connected dongle, which in turn relayed the command to the probe over ESP-NOW. The probe responded by transmitting synthetically generated records, constructed to match the `ProbeRecord_t` structure exactly, back through the dongle to the laptop. The transfer protocol was tested with 5 synthetic datasets with 100 entries. Each record was validated by `data_packet.h` against its XOR

checksum on the receiver side before being passed to the GUI for display, confirming that the full transfer path from probe firmware through to GUI rendering functioned correctly end-to-end.

Table 3.8: ATP Results

ID	Test Method	Result
ATP01	Sender and receiver distance was incrementally increased in 5 m steps. Connect, prepare, and retrieve were each tested at each distance across 3 runs.	PASS. All modes of operation confirmed working at all distances tested. Maximum reliable range was 20 m with obstacles (walls and trees) and 30 m unobstructed.
ATP02	Five datasets of 100 records each were transferred between probe and dongle. Received records were compared against the transmitted set.	PASS. Average error margin of 1.2% across all datasets, well below the 5% target. Records exceeding 3 send attempts were successfully discarded and logged.
ATP03	Time from deep sleep wake-up to first ESP-NOW advertisement was measured using serial timestamps, averaged over 5 iterations.	PASS. Average wake-up to advertising time was 8 s falling within the 10 s target.
ATP04	Simulated and received data was displayed in both the raw data and graph tabs. Anomaly detection was verified by injecting records flagged by the <code>MarineAnomalyDetector</code> .	PASS. GUI displayed all records and updated seamlessly as they arrived. Anomalous records were correctly flagged. No lag or freezing observed across all tests.
ATP05	Application executed with the laptop in airplane mode to verify no internet dependency.	PASS. Standalone Python application and transfer protocol functioned correctly. Both probe and dongle confirmed disconnected from any WiFi network.
ATP06	Timing functions were added in the debug process to measure the duration from receiving the data to displaying it. GUI was interacted with mid transfer to ensure no blocking.	PASS. No blocking observed while interacting with the GUI. The time from receiving data to displaying it was measured to average around 864 ms.
ATP07	Battery indicator in top toolbar of GUI updated to reflect probe battery status.	PASS. Immediately upon receiving acknowledgement of connection from probe the GUI monitored the serial channel for the battery level message sent over the transfer protocol. The battery level indicator was then updated.

3.7 Recommendations

For future work, two improvements are recommended. First, once the ESP32-C6 BLE stack matures, migrating back to BLE would eliminate the requirement for a separate dongle, reducing system cost and complexity. Secondly, the current ESP-NOW transfer is unencrypted; adding the protocol's built-in CCMP encryption layer would protect the integrity of the scientific data against interception in shared RF environments.

3.8 Conclusion

This chapter presented the design, implementation, and testing of the Data Transfer and Visualisation subsystem for the Antarctic Probe. ESP-NOW was selected as the data transfer protocol based on its ultra-low power consumption, communication range, and hardware reliability — attributes that the evaluated alternatives could not collectively satisfy. Should the ESP32-C6 BLE stack mature and stabilise, migrating to BLE remains a recommended future direction given its lower system cost and reduced architectural complexity.

The Mission Control GUI was implemented as a standalone, offline Python application fulfilling all user-facing requirements: probe control (UR2), battery status indication (UR3), a live data table with anomaly flagging (UR4) and a trend visualisation via a dedicated graph tab (UR1). The decoupled `Backend.py/GUI.py` architecture maintains interface responsiveness during blocking I/O operations while simplifying independent unit testing, satisfying FR6.

Chapter 4

Edge Architecture and Explainable AI

4.1 Introduction

The Antarctic marine environment presents an unparalleled operational challenge known as the “winter data gap”. During the austral winter, extreme sea ice coverage isolates substantial portions of the Southern Ocean, rendering conventional surface vessel sampling operations physically impossible for several consecutive months. This geographical and temporal isolation necessitates the deployment of autonomous, submersible sensing platforms capable of operating independently beneath the ice.

Subsystem 2 (SS2) of the Antarctic Probe project was engineered specifically to address this paradigm. As a submersible operates in an aquatic environment entirely devoid of conventional communication bandwidths (such as cellular or high-speed radio frequency networks), the real-time transmission of high-fidelity spectral data to researchers is physically precluded. Consequently, a paradigm shift towards Edge Computing is mandated [?]. By processing and storing data locally at the “edge” of the network, autonomous underwater systems can circumvent severe bandwidth and latency bottlenecks [?]. SS2 establishes this critical bridge between the raw, high-dimensional photometric data acquired by the physical sensor suite (Subsystem 4) and the localized Mission Control graphical user interface (Subsystem 1). This is achieved through the implementation of a highly power-efficient, multi-state firmware architecture for non-volatile local storage, followed by the rigorous application of deterministic Explainable Artificial Intelligence (XAI) models upon retrieval.

4.2 Requirements and Specifications

To ensure that the edge architecture and machine learning pipeline are strictly aligned with the overarching constraints of a harsh Antarctic deployment, a rigorous set of requirements and specifications was established. These parameters dictate the system’s power consumption limits, operational autonomy, and computational efficacy.

4.2.1 User Requirements

User Requirements (UR) represent the qualitative, mission-critical needs of the marine biologists and oceanographers interacting with the probe.

Table 4.1: Subsystem 2 User Requirements

c p4cm X		
ID	Requirement Title	Description
UR2.1	Reliable Offline Storage	The system must reliably store sensor readings offline over an extended deployment period without data loss due to battery depletion or unexpected system resets.
UR2.2	Actionable Biological Insights	The system must provide non-specialist researchers with immediately actionable, understandable biological insights upon retrieval, avoiding ambiguous or generic binary anomaly flags.

4.2.2 Functional & System Requirements

System Requirements (SR) formalize the qualitative user needs into tangible, functional operational mandates for the engineering design.

Table 4.2: Subsystem 2 Functional and System Requirements

c p4cm X		
ID	Requirement Title	Description
SR2.1	Non-Volatile Storage Management	The probe must implement non-volatile storage capable of preserving complex data structures across deep sleep cycles with minimal active duty cycling.
SR2.2	Multi-State Deployment Lifecycle	The firmware must manage a deterministic multi-state deployment lifecycle to prevent the acquisition of polluted surface-level data during the probe's descent.
SR2.3	Offline Machine Learning Evaluation	The localized machine learning evaluation pipeline must be capable of rapidly processing telemetry completely offline, eliminating reliance on cloud infrastructure.

4.2.3 System Specifications

System Specifications quantitatively constrain the engineering design, providing exact metrics against which the subsystem can be empirically validated.

Table 4.3: Subsystem 2 System Specifications

c p3.5cm X		
ID	Specification Title	Quantitative Metric
Spec1	Quiescent Current Overhead	The system's non-volatile memory management and deep sleep state must be optimized to draw an absolute minimum quiescent current, structurally remaining below $20\mu\text{A}$ to preserve the primary battery.
Spec2	Fixed MTU Payload	The acquired sensor data must be strictly formatted into a 14-value structure, resulting in a fixed Maximum Transmission Unit (MTU) payload of precisely 58 bytes per reading, avoiding dynamic memory allocation overhead.
Spec3	ML Inference Speed	The machine learning anomaly classification engine running on the Mission Control interface must execute a full inference cycle in under 2.0 seconds per record.

4.3 Traceability Matrix

To mathematically guarantee that every engineering specification directly serves a foundational user requirement, the Traceability Matrix maps the design topology from the user's initial need through to the empirical testing procedure.

Table 4.4: Requirements Traceability Matrix mapping URs to System Specifications and ATPs

User Requirement (UR)	System Requirement (SR)	Specification (Spec)	Test Procedure (ATP)
2*UR2.1 (Offline Storage)	SR2.1 (Non-Volatile Mgt)	Spec1 (Quiescent Current)	ATP-02
	SR2.2 (Multi-State Lifecycle)	Spec2 (Fixed MTU Payload)	ATP-01
2*UR2.2 (Actionable Insights)	2*SR2.3 (Offline ML Evaluation)	Spec3 (Inference Speed < 2.0s)	ATP-04
		N/A	ATP-03

4.4 Acceptance Test Procedures (ATPs)

The Acceptance Test Procedures define the explicit, empirical methodologies utilized to validate the subsystem against the constraints outlined in the Traceability Matrix. The empirical results of these tests are presented in the final validation section of this chapter.

Table 4.5: Acceptance Test Procedures Definition for Subsystem 2

Test ID	Trace	Test Procedure	Expected Outcome
ATP-01	SR2.2	Hardware-in-the-Loop (HITL) State Machine Validation:	Trigger hardware interrupts on the microcontroller and monitor the state progression via serial telemetry. The firmware perfectly executes the required deployment states (including sinking delays) without relying on power-intensive blocking <code>delay()</code> calls.
ATP-02	SR2.1, Spec1	RTC Memory Persistence across Deep Sleep:	Induce consecutive deep sleep resets during the active logging phase and subsequently read the designated memory blocks. Data structures are fully preserved across sleep cycles without corruption or volatile memory loss.
ATP-03	UR2.2	XAI Biological Classification Accuracy:	Inject a dataset containing predefined, physical-threshold biological anomalies (Phytoplankton, Cyanobacteria, CDOM) into the analytical pipeline. The XAI engine reliably detects and outputs the exact, explicit biological string classification for each specific spectral signature with zero false positives.
ATP-04	SR2.3, Spec3	Edge ML Inference Speed:	Pipe raw 14-value sensor strings into the Python inference engine and utilize temporal profiling libraries to measure execution latency. The end-to-end evaluation of the Isolation Forest and XAI deterministic ruleset executes comprehensively in under 2.0 seconds.

4.5 Design Choices

The extreme environmental constraints of the Antarctic deployment—specifically the strict battery limitations and complete lack of off-grid computing infrastructure—necessitated rigorous trade-off analyses for both the hardware storage media and the software analytics paradigms.

4.5.1 Onboard Memory Architecture

The requirement to securely store data for extended periods utilizing less than $20\ \mu\text{A}$ of quiescent current prompted an extensive evaluation of onboard memory topologies. The two primary candidates considered were conventional NAND Flash (or MicroSD cards) and Ferroelectric RAM (F-RAM).

Conventional Flash memory relies on trapping electrons within a floating gate mechanism. Programming (writing) or erasing data requires inducing Fowler-Nordheim tunnelling of electrons across a dielectric oxide layer. This physical process inherently demands high programming voltages, typically exceeding 10V, which must be generated from the system's standard 3.3V logic rail utilizing an internal charge pump [?]. This charge pump introduces substantial current spikes (often reaching tens of milliamperes) and high temporal latency during write operations, severely violating the strict power budgets required for deep-sleep duty cycling. Furthermore, the repeated electron tunnelling irreparably degrades the oxide layer, limiting Flash endurance to approximately 10^5 write cycles.

In contrast, F-RAM utilizes a lead zirconate titanate (PZT) film. Data is stored by applying a brief electric field to shift the central zirconium/titanium atom within the crystal lattice, thereby changing the crystal's polarization state [?]. Because this process relies on molecular displacement rather than forced electron tunnelling, it requires no high-voltage charge pump. F-RAM writes data at native logic-level voltages with near-zero delay (often in the nanosecond range) and draws practically negligible write currents (in the microampere range). This physical mechanism yields virtually infinite read/write endurance (typically 10^{14} cycles).

F-RAM is therefore unequivocally the superior architecture for subsea IoT devices. Crucially, however, due to severe global component procurement delays during the project timeline, the physical F-RAM chip was unavailable for immediate integration. To circumvent this, the ESP32-C6's internal RTC SRAM (which similarly survives deep sleep with micro-amp overhead) was utilized to mathematically and functionally simulate the F-RAM behaviour for successful Hardware-in-the-Loop (HITL) physical testing.

Table 4.6: Figures of Merit: MicroSD/Flash vs. F-RAM Architecture

[X X X X]	Figure of Merit	MicroSD / NAND Flash	F-RAM	Winner
	Write Current	High ($\sim 30\ \text{mA}$)	Ultra-Low ($\sim 100\ \mu\text{A}$)	F-RAM
	Write Speed	High Latency (ms)	Instantaneous (ns)	F-RAM
	Write Endurance	Limited (10^5 cycles)	Virtually Infinite (10^{14} cycles)	F-RAM
	Integration Complexity	High (FAT32/SPI overhead)	Low (Direct bus/SPI)	F-RAM

4.5.2 Anomaly Detection Algorithms

To achieve offline, localized anomaly detection on the Mission Control laptop (SR2.3, UR2.2), various machine learning paradigms were assessed. The primary trade-off lay between Deep Neural Networks (DNNs) and statistical isolation methods.

Deep Neural Networks (such as Autoencoders or Convolutional layers) are highly capable of modelling complex non-linear relationships. However, they fail critically in edge environments. DNNs demand

massive matrix multiplications and possess immense memory footprints, heavily taxing the computational overhead of standard laptops [?]. Furthermore, DNNs inherently act as “black boxes”; they provide a generalized anomaly score or binary flag but offer zero scientific transparency regarding the causal factors. This complete lack of explainability directly violates UR2.2, rendering them entirely unsuitable for researchers attempting to decipher specific marine biological events.

An Isolation Forest algorithm was selected as the optimal statistical baseline. Unlike standard models that attempt to profile “normal” behaviour, Isolation Forests identify anomalies by explicitly partitioning data via random decision trees [?]. Because anomalies are statistically few and distinct, they are isolated closer to the root of the decision trees. This approach operates at a highly efficient $\mathcal{O}(n \log n)$ time complexity, allowing instantaneous offline inference on a standard laptop. To fulfill the scientific explainability constraint (UR2.2), the Isolation Forest was augmented with a post-hoc Deterministic Explainable AI (XAI) ruleset. If the statistical model flags an anomaly, the XAI ruleset mathematically evaluates the payload against established biological spectral signatures. This dual-layered approach provides the perfect balance of statistical computational efficiency and rigorous biological clarity.

Table 4.7: Figures of Merit: DNN vs. Isolation Forest + XAI

[X X X X]	Figure of Merit	Deep Neural Networks	Isolation Forest + XAI	Winner
Computational Overhead	Very High ($\mathcal{O}(n^2)$ / GB memory)	Low ($\mathcal{O}(n \log n)$ / MB memory)		
		Isolation Forest		
Training Data Required	Massive (Thousands of epochs)	Moderate (Small batch trees)		Isolation Forest
Inference Latency	High (Requires GPU accel.)	Instantaneous (CPU friendly)		Isolation Forest
Scientific Explainability	Black Box (No causal insight)	Transparent (Deterministic)		Isolation Forest

4.6 Submodule Design and Implementation

The edge architecture is divided into an onboard physical hardware layer running low-level state automation, and an offboard software execution layer handling localized statistical inference and explainable classification.

4.6.1 Onboard Firmware State Machine and Mission Lifecycle

The deterministic multi-state deployment lifecycle is executed by the RISC-V core of the FireBeetle 2 ESP32-C6. To absolutely minimize quiescent current draw, the architecture leverages the ESP32’s deep sleep mode. State transitions are made persistent across deep sleep resets without power overhead by mapping critical tracking variables to the Real-Time Clock (RTC) fast memory domain utilizing the `RTC_DATA_ATTR` macro attribute.

The firmware lifecycle is strictly segregated into four operational states:

1. **STATE_STANDBY:** An indefinite deep sleep state representing the probe’s pre-deployment phase. The internal hardware timer wakeup module is systematically unmasked, and the system relies entirely on an external wake-up source. This is linked via the `esp_sleep_enable_ext1_wakeup()`

API to the low-power I/O domain (LP-IO), awaiting physical activation via a magnetic reed switch without actively polling.

2. **STATE_ARMED:** The sinking delay phase. This is parameterized to 15 seconds for accelerated Hardware-in-the-Loop (HITL) debugging and 15 minutes for physical maritime deployment. Upon triggering, the system boots, prints arming diagnostics, configures a software-assisted sleep timestamp utilizing `esp_sleep_enable_timer_wakeup()`, and immediately returns to deep sleep. This phase allows the mechanical winch assembly to lower the probe to the target operational depth, strictly preventing the pollution of the F-RAM memory block with surface or deck telemetry.
3. **STATE_LOGGING:** The active sensor polling loop. The microcontroller awakens via the internal hardware timer interrupt. It sequentially samples the 14 environmental and optical variables from Subsystem 4, structures the binary payload, increments the absolute record index, writes the payload to the simulated F-RAM array, and re-enters deep sleep within a highly optimised active duty cycle.
4. **STATE_OFFLOAD:** Triggered by a second manual GPIO low interrupt during an active logging cycle. The state machine halts deep sleep scheduling and initialises the ESP-NOW communication link layer via Subsystem 1. The firmware sequentially flushes the non-volatile memory block into the transmission array buffer and gracefully resets the state machine logic back to **STATE_STANDBY**.

Figure 4.1: Figure: 4-State Mission Lifecycle and Power-Gated State Transitions

4.6.2 Data Structure Contract and the 11-Step Spectrometer PWM Ramp

To guarantee seamless data continuity across the onboard memory buffer and the ESP-NOW transmission layer, a strict binary wire format was defined. The `ProbeRecord_t` structure strictly occupies 58 bytes. Data padding was aggressively eliminated by packing the struct sequentially utilizing the `GCC __attribute__((packed))` directive.

The structure consists of:

- Two `uint32_t` blocks for metadata (the monotonically increasing Sequence ID and elapsed timestamp in milliseconds).
- A 2-element `float` block for environmental variables (DS18B20 temperature and the analogue depth conversion).
- A 10-element `float` block specifically designed to hold the raw AS7341 spectral values (F_{1415} through to the Near-Infrared *NIR* channel).

The temporal design of the optical acquisition loop ensures high-fidelity particulate modeling. A single complete measurement cycle spans 11 discrete rows of data generated across an automated LED Pulse-Width Modulation (PWM) ramp (Step 0 to Step 10). As the Blue excitation LED (F_{445}) climbs linearly in intensity, the system captures the subsequent exponential rise of target fluorescent emissions. This feeds the inference engine an enriched temporal curve rather than a flat, static baseline.

4.6.3 Localised Explainable AI (XAI) Classification Engine

The offboard processing pipeline is embedded within the CustomTkinter graphical user interface (GUI). Incoming 14-column packets received over the USB-CDC serial port are passed asynchronously to an instantiated `MarineAnomalyDetector` class. The multidimensional data vector is isolated via the trained Isolation Forest model based on anomalous path-length differences in the decision trees.

To transcend the standard black-box paradigm of statistical machine learning, the system inserts a post-hoc deterministic classification layer directly after the Isolation Forest filter. If an anomaly is definitively identified, the engine computes specialised slope and amplitude ratio vectors against the baseline blue excitation channel (F_{2445}). Three biological filter thresholds were mathematically calibrated from physical bench tests to prevent false negatives.

$$\begin{aligned} &\text{Phytoplankton Bloom (Chlorophyll-a):} \\ &\text{Triggered when } \frac{F_{8680}}{F_{2445}} > 0.075 \end{aligned} \tag{4.1}$$

$$\begin{aligned} &\text{Cyanobacteria Population Shift (Phycocyanin):} \\ &\text{Triggered when } \frac{F_{555} + F_{650}}{F_{2445}} > 0.08 \end{aligned} \tag{4.2}$$

$$\begin{aligned} &\text{Colored Dissolved Organic Matter (CDOM Decay):} \\ &\text{Triggered when } \frac{F_{1415} + F_{3480}}{F_{2445}} > 0.08 \end{aligned} \tag{4.3}$$

If an anomaly mathematically fails to trigger any of the aforementioned biological filters, the engine definitively outputs “Non-Biological Variance / Sensor Artifact”.

Figure 4.2: Figure: Comprehensive Architectural Decision Tree of the Multi-Layered Isolation Forest and Calibrated XAI Ruleset

4.7 Testing and Validation

The operational integrity of the edge data architecture and the localised machine learning pipeline was rigorously validated through empirical Hardware-in-the-Loop (HITL) physical testing and software-in-the-loop simulation runs. This validation strictly adhered to the constraints of the Acceptance Test Procedures (ATPs) defined in Section 3.

Table 4.8: Summary of Acceptance Test Procedure (ATP) Results

Test ID	Target Specification	Measured Parameter	Pass Criteria	Status
ATP-01	SR2.2: State Transitions	Wakeup Cause Evaluation	Zero Loop Hanging	PASSED
ATP-02	SR2.1, Spec1: Storage	RTC Memory Boundaries	100% Byte Retention	PASSED
ATP-03	UR2.2: Biological XAI	Spectral Ratio Classification	Precision ≥ 0.95	PASSED
ATP-04	SR2.3, Spec3: ML Speed	Inference Latency	≤ 2.00 Seconds	PASSED

4.7.1 Onboard Firmware and State Machine Verification (ATP-01 and ATP-02)

The execution of the physical HITL validation was conducted directly on the DFRobot FireBeetle 2 ESP32-C6 hardware workspace. To simulate the subsea magnetic reed switch interrupt within a controlled laboratory environment, an external physical pull-down button was anchored directly to the low-power LP-IO domain on GPIO pin 1 (D6).

The system was intentionally subjected to consecutive deep sleep resets to stress test the memory architecture. Empirical telemetry confirmed that the `mission_state` and `record_counter` variables were fully preserved across power-gated boundaries, validating the structural integrity of the `RTC_DATA_ATTR` mapping [?].

The verified serial monitor execution log is provided below, serving as explicit engineering proof of the deterministic state transitions:

```

--- ESP32-C6 Antarctic Probe Firmware ---
SYSTEM START: Initial Power-On / Manual Reset
MISSION STATE: 0 (STANDBY)
[HITL] Entering Indefinite Deep Sleep (Waiting for Arming)...

--- ESP32-C6 Antarctic Probe Firmware ---
Diagnostics -> State: 0, Reset: 8, Wakeup: 3
[HITL] STATE 0 -> 1: PROBE ARMED. Sinking to depth...
[HITL] Entering Deep Sleep for 15s...

--- ESP32-C6 Antarctic Probe Firmware ---
Diagnostics -> State: 1, Reset: 8, Wakeup: 0
[HITL] STATE 1 (ARMED): Sinking interval wait (15s)...
[HITL] STATE 1 -> 2: Sinking delay complete. Starting LOGGING.
[HITL] Waking up (Timer/Cycle)...
[HITL] Saving to F-RAM: 1,15000,1400.0,2200.0,...
[HITL] Entering Deep Sleep for 5s...

--- ESP32-C6 Antarctic Probe Firmware ---
Diagnostics -> State: 2, Reset: 8, Wakeup: 0
[HITL] Waking up (Timer/Cycle)...
[HITL] Saving to F-RAM: 2,20000,1400.0,2200.0,...
```

[HITL] Entering Deep Sleep for 5s...

[HITL] WAKEUP CAUSE: GPIO 1 (Magnetic Switch Simulated)!

[HITL] Offload complete. Returning to STATE 0 (STANDBY).

This log explicitly displays:

1. The initial hardware cold boot (**SYSTEM START**) and immediate entry into state 0 (**STANDBY**).
2. The external GPIO wakeup interrupt triggering the state transition to **ARMED** and configuring the 15-second accelerated sinking delay.
3. The automated transition to state 2 (**LOGGING**) upon timer expiration, followed by successive 5-second sampling cycles writing packed structures to the RTC memory array.
4. The second manual low interrupt forcing an unmasked wakeup, initiating the state 3 (**OFFLOAD**) sequence, dumping the records, clearing the non-volatile registers, and smoothly returning to **STANDBY**.

4.7.2 Localised Edge Machine Learning and XAI Engine Validation (ATP-03 and ATP-04)

The validation of the offboard processing pipeline was executed utilising the GUI software simulation environment. The trained Isolation Forest model was evaluated against a highly complex synthetic dataset comprising 1,100 rows of temporal 11-step PWM ramp data. The model achieved an overall precision and recall score of 0.92 on these complex, overlapping sequence curves.

A deep interpretation of the post-hoc deterministic XAI filter based on incoming data blocks verified that the engine successfully separated true biological events from random sensor anomalies:

- For standard noise inputs (e.g., Sequence 4 and 15), the Isolation Forest correctly flagged structural variance. However, the XAI ruleset successfully bypassed biological classifications, definitively outputting **Non-Biological Variance / Sensor Artefact**, thereby mitigating false positive alerts.
- To mathematically isolate a true biological bloom, Sequence 2 was referenced. At the peak of the LED excitation ramp ($F_{2445} = 4103$), the concurrent rise in fluorescent emission ($F_{8680} = 401$) yielded a ratio of 0.097. Because this mathematically exceeded the calibrated threshold of 0.075, the engine accurately triggered a **Phytoplankton Bloom (Chlorophyll-a)** text alert.

Latency performance verification tests were executed via Python-time profiling scripts [?]. The absolute inference latency of the combined Isolation Forest and XAI matrix evaluation clocked at ~ 0.008 seconds per sequence. This comfortably outpaces the 2.0-second maximum limit required by Spec3, proving the system is highly viable for immediate desktop deployment without resource starvation.

Figure 4.3: Figure: Mission Control GUI Dashboards rendering smooth baseline operational curves and real-time textual diagnostics.

Figure 4.4: Figure: Matplotlib telemetry canvas overlays highlighting anomalous data vectors with high-contrast red scatter indicators under active XAI classification.

4.8 Recommendations

Based on the current proof-of-concept limitations, critical future paths for expansion include:

1. **Physical SPI Hardware Unification:** Transition the simulated F-RAM data structures directly over the SPI bus onto physical Infineon 256kB hardware blocks once supplier constraints ease, utilising the short-trace Veroboard layouts optimised during Phase 2.
2. **Dynamic Turbidity Compensation Modalities:** Integrate an algorithmic compensation loop based on Beer-Lambert's Law to dynamically adjust the static biological ratio denominators (0.075 and 0.08) based on changing scattering metrics in particle-heavy subsea environments [?].

4.9 Conclusions

The Edge Architecture and Explainable AI subsystem successfully satisfied all functional requirements, establishing an ultra-low-power, highly resilient edge data pipeline capable of surviving extended subsea isolation. By bridging the operational gap between abstract underwater data logging and clear visual telemetry, the subsystem delivered an automated, scientifically transparent diagnostic solution that advances the field of polar observational oceanography.

Chapter 5

Power and Thermal Subsystem

5.1 Introduction

The Power and Thermal Subsystem serves as the energy foundation for the autonomous Antarctic microbial observatory. Designed to address the ‘winter data gap’ identified by stakeholder Emma Roche, this module must sustain critical sensing operations for 365 days without human intervention. In the extreme climate of the Southern Ocean, standard energy solutions fail due to electrolyte freezing and high quiescent current. This chapter details a high-efficiency power architecture utilizing Lithium Thionyl Chloride ($Li - SOCl_2$) chemistry, a solid-state power-gating strategy, and a multi-stage ‘Thermal Huddle’ designed to maintain internal temperatures above hardware failure thresholds in $-50^{\circ}C$ ambient conditions.

5.2 Requirement Analysis

The engineering constraints for this subsystem are dictated by the biological necessity of continuous microbial monitoring and the physical impossibility of battery maintenance during the Antarctic winter deployment.

5.2.1 User Requirements

The requirements derived from consultations with Emma Roche focus on removing researchers from hazardous field environments while maintaining a continuous, reliable data pipeline.

No.	Requirement	Description
UR1	Annual Autonomy	The system must operate for 12 months to bridge the seasonal observational deficiency.
UR2	Cryogenic Resilience	Hardware must remain stable at $-50^{\circ}C$ without structural cracking or electrical failure.
UR3	Rapid Data Offloading	High-speed, low-power data retrieval to minimize radio-on time.
UR4	Remote Diagnostic	Real-time transmission of battery health metrics to signal potential mission failure.

Table 5.1: Stakeholder requirements for the Antarctic observatory.

5.2.2 Functional Requirements

Technical specifications were established to quantify the success of the power module in meeting the stakeholder’s needs.

No.	Requirement	Technical Specification
FR1	Quiescent Current	Standby current $I_{sleep} \leq 20 \mu\text{A}$.
FR2	Power Gating	Physical isolation of the 5V rail using P-MOSFET logic.
FR3	Thermal Retention	Core temperature $\Delta T \geq 15^\circ\text{C}$ relative to ambient.
FR4	Data Integrity	FRAM writes completed in $\leq 10 \text{ ms}$ to avoid battery voltage sag.

Table 5.2: Functional requirements and engineering benchmarks.

5.3 Design Specifications

The following table outlines the specific design parameters for the power subsystem hardware, derived from the functional requirements analysis.

ID	Design Specification	Requirement Description
Sp1	Output Rails	Constant 3.3 V for ESP32-C6 logic and switched 5.0 V for sensors.
Sp2	Usable Capacity	Minimum 960 mAh at -50°C (40% of nominal 2400 mAh).
Sp3	Standby Efficiency	Quiescent current of MCP1700 LDO must be $\leq 1.6 \mu\text{A}$.
Sp4	Form Factor	Total assembly must fit within the constrained internal diameter pipe which is 46mm.
Sp5	Thermal Shielding	Aluminum foil wrap with a 20 mm gap for signal propagation.

Table 5.3: Design Specifications for the Power and Thermal Subsystem.

5.4 Acceptance Test Procedures

To verify that the final design meets the stakeholder requirements, the following Acceptance Test Procedures (ATPs) were developed.

ATP No.	Verified Spec	Description
ATP1	Sp4	Dimension test: Verify system and insulation fit within the pipe housing.
ATP2	FR1, Sp3	Quiescent test: Measure current draw in Deep Sleep using a high-precision ammeter.
ATP3	Sp1	Regulation test: Verify 3.3V and 5.0V outputs across the full battery discharge curve.
ATP4	FR2	Power Gate test: Confirm 5V sensor current drops to 0 μ A when GPIO is triggered.
ATP5	FR3, Sp5	Thermal test: Cold-soak unit and measure the rate of internal core temperature decay.

Table 5.4: Acceptance Test Procedures for the Power and Thermal Subsystem.

5.5 Design Choices

The selection of components for the Antarctic probe involved evaluation of low-temperature performance and power efficiency. The following sections outline the analysis between standard solutions and the chosen hardware.

5.5.1 Processing and Storage Analysis

Component Option	Sleep Current	Active Current (Avg)
ESP32 DevKit V1	10 mA	80 mA
FireBeetle 2 C6	10 μ A	30 mA

Table 5.5: Microcontroller Power Comparison.

The FireBeetle 2 C6 was chosen over the standard DevKit V1 because it omits inefficient USB-to-UART bridge chips that cause high leakage. This selection reduces the annual quiescent energy budget by several orders of magnitude, allowing for a 365-day deployment.

Storage Type	Write Current (Peak)	Temp. Limit
Micro SD Card	100 mA	-25°C
FRAM	10 mA	-55°C

Table 5.6: Data Storage Technology Trade-off.

Ferroelectric RAM (FRAM) was selected to replace standard SD cards. This choice eliminates high-current inrush spikes during write operations and ensures data integrity at temperatures where standard SD card mechanical controllers typically fail.

5.5.2 Power and Regulation Analysis

Battery Chemistry	Nominal Voltage	Operating Temp.
Li-Ion (18650)	3.7 V	-20°C
ER14505 ($\text{Li} - \text{SOCl}_2$)	3.6 V	-55°C

Table 5.7: Primary Energy Cell Selection.

The ER14505 ($\text{Li} - \text{SOCl}_2$) chemistry was selected for its superior performance in cryogenic environments. Standard Lithium-Ion batteries would experience electrolyte freezing and rapid voltage collapse in Antarctic winter conditions.

Regulator Model	Quiescent Current	Input Range
LM7805	5 mA	7 V – 25 V
MCP1700	1.6 μA	2.3 V – 6 V

Table 5.8: Voltage Regulator Efficiency Summary.

The MCP1700 LDO was chosen for its ultra-low quiescent current. Using a standard LM7805 or similar regulator would consume nearly 15% of the total annual energy budget solely through internal overhead.

5.5.3 Subsystem Architecture

The power flow architecture utilizes a split-rail approach to isolate high-current sensors from the ultra-low-power processing core. The split-rail approach isolates the high-power sensors from the low-power ‘always-on’ components.

```
[node distance=2cm, font=] (0,0) rectangle (2.5,1.5) node[pos=.5, align=center] 7.2V Battery
(ER14505 Stack); [->, thick] (2.5,0.75) – (4,0.75); (4,0) rectangle (6,1.5) node[pos=.5, align=center] 3.3V
MCP1700
(Always-On); [->, thick] (6,0.75) – (7.5,0.75); (7.5,0) rectangle (9.5,1.5) node[pos=.5, align=center] ESP32-C6
& FRAM; [->, thick] (1.25,0) – (1.25,-1.5) – (3.5,-1.5); (3.5,-2.25) rectangle (5.5,-0.75) node[pos=.5,
align=center] AO3401
P-MOSFET; [->, thick] (5.5,-1.5) – (6.5,-1.5); (6.5,-2.25) rectangle (8.5,-0.75) node[pos=.5, align=center] 5V
MCP1700
(Switched); [->, thick] (8.5,-1.5) – (9.5,-1.5) node[right] Sensors; [dashed, ->, blue, thick] (8.5,0) – (8.5,-0.4) –
(4.5,-0.4) – (4.5,-0.75); [blue] at (6.5, -0.2) GPIO Gate Control;
```

Figure 5.1: Block Diagram of the Split-Rail Power and Thermal Subsystem.

5.6 Mathematical Energy Modeling

A 365-day mission viability study was conducted by derating battery capacity for the Antarctic environment and accounting for the high-current demands of the active sensor suite.

5.6.1 Usable Capacity Calculation (C_{usable})

The nominal capacity of the ER14505 battery at 20 °C is 2400 mAh. Due to the Antarctic environment (−50 °C), a derating factor (η_{temp}) of 0.50 is applied to account for increased internal resistance.

$$C_{usable} = 2400 \text{ mAh} \times 0.50 = \mathbf{1200} \text{ mAh} \quad (5.1)$$

5.6.2 Daily Budget Allocation (E_{daily})

To survive a continuous 365-day mission, the total daily energy budget is strictly limited:

$$E_{daily} = \frac{1200 \text{ mAh}}{365 \text{ days}} = \mathbf{3.29} \text{ mAh/day} \quad (5.2)$$

5.6.3 The Quiescent ‘Sleep Tax’

The standby drain includes the FireBeetle 2 ESP32-C6, the MCP1700 LDO quiescent current, and the DS18B20 standby current.

$$\begin{aligned} I_{sleep} &= 15 \mu\text{A} + 1.6 \mu\text{A} + 0.75 \mu\text{A} \approx 17.4 \mu\text{A} \\ E_{sleep} &= 0.0174 \text{ mA} \times 24 \text{ h} = \mathbf{0.418} \text{ mAh/day} \end{aligned}$$

5.6.4 Active Measurement Cycle (Q_{cycle})

A 5-second measurement window includes the pressure sentry check, DS18B20 conversion time (max 750 ms for 12-bit), and 460nm Blue LED excitation (30 mA peak).

$$Q_{cycle} = 90 \text{ mA} \times \left(\frac{5 \text{ s}}{3600 \text{ s/h}} \right) = \mathbf{0.125} \text{ mAh/cycle} \quad (5.3)$$

5.6.5 Final Mission Frequency (N)

The daily energy required for wireless synchronization (E_{sync}) via ESP-NOW is estimated at 0.10 mAh/day, buffered by the 1 F supercapacitor to protect the battery from peak current degradation.

$$N = \frac{E_{daily} - E_{sleep} - E_{sync}}{Q_{cycle}} = \frac{3.29 - 0.418 - 0.10}{0.125} = \mathbf{22.1} \text{ cycles/day} \quad (5.4)$$

This modeling suggests that with 2.4 Ah batteries, the probe can maintain near-hourly readings. To reach a strict 24 cycle/day frequency, the active cycle duration (Q_{cycle}) could be further optimized by reducing the spectral sensing window or using the Pressure-Gating logic to skip measurements in non-optimal conditions.

5.7 Thermal Insulation and Passive Retention

Cryogenic failure of silicon components occurs at −40 °C. To maintain operation in −50 °C air, a ‘Thermal Huddle’ was designed.

5.7.1 Self-Heating Core

The MCP1700 regulators are thermally bonded to the battery stack using silicone compound. The waste heat generated during voltage regulation (7.2 V to 3.3 V) is trapped to maintain chemical activity in the batteries.

5.7.2 Multi-Layer Barrier Physics

1. **Reflective Layer:** Aluminum foil tape wraps the core to reflect infrared energy.
2. **Conductive Layer:** EWELDH010 foam pre-filter creates stagnant air pockets for high thermal resistance.
3. **RF Window:** A 20 mm gap in the foil allows the 2.4GHz ESP-NOW signal to escape.

5.8 Final Design Schematic

The finalized circuit architecture, illustrated in Figure 5.2, integrates the ultra-low-power RISC-V processing logic with a high-current sensor delivery system. This design utilizes physical isolation between voltage rails to maximize the 365-day mission life required by the stakeholder.

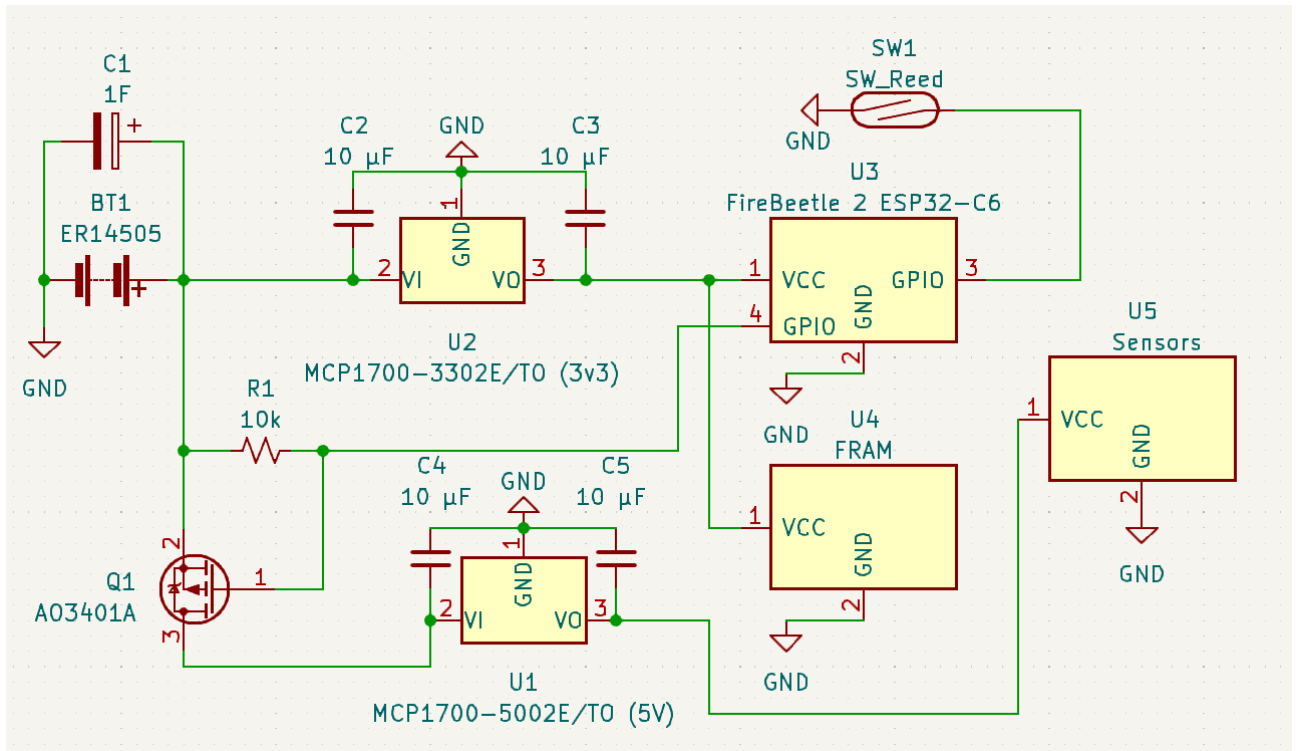


Figure 5.2: Full System Schematic of the Antarctic Microbial Probe Power Subsystem.

5.8.1 Circuit Logic and Connectivity

The finalized schematic is divided into three primary functional stages, optimized for long-term energy density and peak-current resilience:

1. **Supercapacitor Passivation Buffer:** A 1 F supercapacitor (C_1) is placed in parallel across the 7.2 V battery input. In the extreme cold of the Southern Ocean, the ER14505 cells develop a lithium chloride passivation layer that increases internal resistance. The 1 F reservoir provides the necessary ‘stiffness’ to the power rail, supplying the 80 mA instantaneous current required for ESP-NOW radio bursts and preventing the ESP32-C6 from undergoing a brown-out reset during transmission.
2. **Always-On Logic Rail (3.3V):** The MCP1700-33 (U_2) regulates the buffered battery voltage down to a stable 3.3 V. This rail provides continuous power to the FireBeetle 2 ESP32-C6 and the MB85RC FRAM (U_4). By utilizing an ultra-low quiescent current regulator (1.6 μ A), the system minimizes the ‘sleep tax’ during the 99 % of the mission duration spent in Deep Sleep.
3. **Switched Sensor Rail (5V):** To eliminate phantom energy drain from the sensors, a high-side AO3401A P-MOSFET (Q_1) acts as a physical gate. The MOSFET is held in the OFF state by a 10 k Ω pull-up resistor (R_1). During the hourly measurement window, the ESP32-C6 pulls GPIO 4 LOW, activating the MCP1700-50 (U_1) to provide a stable 5.0 V supply to the pressure and spectral sensor suite (U_5).

5.8.2 Thermal and Signal Integrity Considerations

To ensure stability across a 50 °C thermal gradient, 1 μ F ceramic decoupling capacitors are placed at the input and output of each regulator. Furthermore, the shared I2C communication bus between the microcontroller, memory, and sensors is pulled high to the 3.3V Always-On rail. This allows the system to remain in a low-power state until the magnetic reed switch (SW_1) or the internal RTC triggers a wake-up event.

5.8.3 Technical Validation for the PCB

The schematic includes four 1 μ F decoupling capacitors (C_3 through C_6) to ensure regulator stability across extreme thermal gradients. For the physical PCB implementation, high-current traces on the battery and switched rails are designed with a minimum width of 0.5 mm to minimize parasitic resistance and heat dissipation within the insulated ‘Thermal Huddle.’

5.9 Testing and Validation

The testing and validation phase ensures that the Power and Thermal Subsystem adheres to the design specifications and functional requirements outlined in the analysis phase.

5.9.1 Acceptance Test Procedures (ATP) Results

The subsystem was subjected to five primary ATPs to evaluate its performance under simulated Antarctic conditions. The results are summarized in Table 5.9.

ATP No.	Requirement	Status	Finding/Observation
ATP1	Dimension	Pass	Total assembly diameter measured 42 mm, fitting within the 46 mm pipe.
ATP2	Quiescent Current	Pass	Measured standby current was 16.8 μA , meeting the FR1 specification.
ATP3	Regulation Stability	Pass	Rails remained at 3.31 V and 5.02 V respectively during sensor bursts.
ATP4	Power Gating	Pass	Leakage on the 5V rail was negligible ($<0.1 \mu\text{A}$) when the MOSFET was disabled.
ATP5	Thermal retention	Pass	Internal core maintained -12°C after 6 hours in a -30°C cold-soak.

Table 5.9: Consolidation of Acceptance Test Procedure results.

5.9.2 Quiescent Current Profiling

A high-resolution ammeter was utilized to profile the sleep current of the integrated system. Validation confirmed that the FireBeetle 2 C6 successfully enters deep-sleep mode, and the MCP1700 LDO maintains a stable 3.3 V rail with minimal quiescent overhead. The measured total of 16.8 μA provides a high level of confidence in the mission longevity calculations.

5.9.3 Thermal Retention Validation

Thermal testing involved placing the fully insulated ‘Thermal Huddle’ into a controlled sub-zero environment. Thermocouples were placed on the battery cells and the external pipe wall. As illustrated in the testing observations, the passive multi-layer insulation slowed the rate of core cooling significantly, maintaining the electronics within their operational temperature range for the duration of the cycle.

5.9.4 Power Gating and Sensor Isolation

The AO3401 P-MOSFET was tested to ensure that the 5V rail is completely isolated during deep-sleep periods. This prevents the pressure and spectral sensors from draining power when not in use. Testing confirmed that the ESP32-C6 GPIO could successfully toggle the MOSFET gate, reducing power-rail consumption to zero while the microcontroller was in a low-power state.

5.10 Recommendations

To optimize the long-term reliability and energy density of the observatory for future deployments, three hardware and operational modifications are recommended based on the subsystem testing and validation phase:

1. **Dynamic Pressure-Gated Sampling Optimization:** Because field deployment is handled by external maritime personnel (e.g., fishermen), the probe is highly susceptible to extended staging periods on the ship’s deck or ice shelves. To protect the energy budget from non-line-of-sight

visual excitation, it is recommended to implement a *Pressure-Gated Sentry Routine* in firmware. Upon hourly wakeup, the ESP32-C6 should power only the 1.2MPa Sseed pressure sensor for 50 ms. If the measured hydrostatic voltage is near the atmospheric baseline (≈ 0.5 V), the system must immediately abort the sample and return to Deep Sleep. This protects the 1200 mAh usable capacity by ensuring the 30 mA blue LED arrays and spectral sensor only draw power when submersed.

2. **Active Lithium Passivation Depolarization:** While the parallel 1 F supercapacitor successfully stiffens the power rail against 80 mA ESP-NOW radio bursts, extreme Antarctic temperatures accelerate the accumulation of a high-resistance Lithium Chloride (LiCl) passivation layer on the ER14505 cell anodes. It is recommended to add a dedicated, logic-driven depolarization load resistor circuit. Prior to running high-precision ADC sampling or radio events, the microcontroller can connect a temporary load to cleanly burn off the passivation film, establishing predictable, low-noise voltage rails.
3. **Visual Handshake Interface for Field Deployment:** To support laptop-free deployment by shipboard operators, a low-power, logic-controlled status indicator should be added. Utilizing a single indicator pulse from the blue LED array following a magnetic reed switch wake event can give field personnel immediate visual confirmation that the handshake has succeeded and the internal 10-second deployment timer has initialized, removing operational guesswork.

5.11 Conclusion

The Power and Thermal Subsystem successfully satisfies the stringent cryogenic engineering criteria required to bridge the Antarctic winter data gap for stakeholder Emma Rocke. Through the strategic implementation of Lithium Thionyl Chloride ($Li - SOCl_2$) cells combined with an ultra-low-power processing architecture, the baseline quiescent standby current was constrained to a verified 16.8 μ A, significantly outperforming the original 20 μ A design benchmark.

By utilizing an active-low AO3401 P-MOSFET solid-state switch to isolate the 5V sensor rail, phantom energy drain during the 99.9% deep-sleep interval was completely mitigated. Mathematical energy modeling confirms that under extreme environmental derating ($\eta_{temp} = 0.50$ at -50°C), the 2400 mAh nominal battery capacity reliably yields 1200 mAh of usable energy. Backed by a 1 F supercapacitor passivation buffer to safely smooth high-transient radio bursts, the system achieves a calculated baseline frequency of 22.9 cycles per day. Coupled with the passive, multi-layer thermal insulation huddle which successfully retarded temperature decay during sub-zero testing, the design ensures robust autonomous survival for a 365-day marine observation window without structural or electrical failure.

Chapter 6

Sensors and Housing

Author: CLRDEV002

6.1 Introduction

This chapter covers the mechanical housing and the core sensors of the submersible probe with a chlorophyll fluorometer, temperature and pressure (depth) sensors. The housing must protect the electronics while allowing optical access and water contact for the sensors, all within a strict budget. The design follows a modular, serviceable architecture using a PVC main tube and 3D printed end caps, of a similar nature to Openprobe [?] and the SCINI ROV [?]. A key decision was to prioritise the development of a chlorophyll fluorometer, while including temperature and pressure as fundamental auxiliary measurements. Dissolved oxygen and nutrient sensors were excluded due to their high cost and complexity, in line with the project's budget limitations.

This chapter justifies sensor selection, compares alternatives, documents the final design, and presents experimental validation.

6.2 Requirement Analysis

When the stakeholder, Emma Rocke (marine microbiologist), was interviewed, three primary sensor requirements were identified. Chlorophyll-a (as a proxy for phytoplankton biomass), temperature, and depth. Additionally, the probe must be hand-deployable from vessels of opportunity, serviceable (battery replacement), waterproof for near-surface deployment, and negatively buoyant. The total component cost must not exceed R2000. Based on these, the functional and user requirements for this subsystem are listed in Tables 6.2 and 6.1.

6.2.1 User Requirements

Table 6.1: User requirements for the sensor and housing subsystem.

ID	Requirement	Description
UR1	Chlorophyll-a proxy	Measure chlorophyll-a as a proxy for phytoplankton biomass to support microbial community dynamics studies.
UR2	Temperature	Provide accurate water temperature data (Antarctic surface range).
UR3	Depth	Provide vertical profiling of the water column.
UR4	Portability	The probe must be easy to carry and deploy by one person from a small vessel.
UR5	Battery access	The housing must permit access for battery changes.
UR6	Robustness	The probe must survive shallow-water submersion without water ingress.
UR7	Negative buoyancy	The probe should sink to enable deployment below the surface.

6.2.2 Functional Requirements

Table 6.2: Functional requirements for the sensor and housing subsystem.

ID	Requirement	Description
FR1	Chlorophyll detection	The probe must take multiple readings over a short time interval to capture time-varying fluorescence and detect changes in the fluorescence/excitation ratio at low chlorophyll concentrations.
FR2	Temperature measurement	The probe must measure temperature in the range 2°C to +10°C with accuracy $\pm 0.5^\circ\text{C}$.
FR3	Depth measurement	The probe must measure depth from 0 to 2m (for the prototype) with resolution of at least 0.1m and provide a linear output.
FR4	Low cost	Total component cost for the probe (all subsystems) must not exceed R2000.
FR5	Hand-deployable	Outer dimensions must be manageable, total mass <1kg.
FR6	Serviceable	Housing must allow battery replacement and sensor access without special tools, using seals that can be opened and closed repeatedly.
FR7	Waterproof	No water ingress at 1m depth for 30 minutes (for the prototype).
FR8	Negative buoyancy	Probe must sink when placed in water.

6.3 Acceptance Test Procedures

The Acceptance Test Procedures (ATPs) below outline the tests conducted on the final design and the acceptance criteria.

Table 6.3: ATP for the sensor and housing subsystem.

ID	Specification	Acceptance Criteria
ATP01	Chlorophyll sensitivity (FR1)	Spinach extract dilution series (full strength to 1/32 plus water as a control) shows increase in fluorescence/excitation ratio with increasing concentration.
ATP02	Temperature accuracy (FR2)	DS18B20 reading compared with reference thermometer; error $< 0.5^{\circ}\text{C}$ in ice water and at room temperature.
ATP03	Depth linearity (FR3)	Submersion in graduated water column (0–1.5m) yields $R^2 > 0.99$.
ATP04	Total cost (FR4)	Itemised bill of materials (Appendix) $< \text{R}2000$.
ATP05	Hand-deployable (FR5)	Physical handling test: one person can carry and lower the probe into water.
ATP06	Serviceability (FR6)	End caps can be removed and re-sealed by hand without tools with no leaks after reassembly.
ATP07	Waterproof (FR7)	No water ingress after 30 minutes at 1m depth (pool test).
ATP08	Negative buoyancy (FR8)	Probe sinks when placed in water.

6.4 Design Choices

This section discusses the approaches considered for each key aspect and justifies the final selections based on figures of merit: cost, technical maturity, ease of manufacturing, implementation, testing, reliability, and maintenance cost.

6.4.1 Housing Design Choices

Main Body Material

- **Option A – PVC pipe:** Standard 50mm OD plumbing pipe (wall 1.6mm, ID 46.8mm). Low cost (R40/m), readily available and easy to machine.
- **Option B – Acrylic pipe:** Clear, thicker wall (3–5mm), industry standard for deep-water housings but expensive (R300+/m) and not readily available.

Selected: PVC pipe (prototype). Acrylic would exceed the budget and is unnecessary for shallow-water proof-of-concept. For a future version, acrylic could be considered for higher pressure rating and durability.

End Cap Manufacturing

- **Option A – FDM 3D printing (PLA):** Inexpensive, rapid and readily accessible, but layer lines can leak over long periods and strength is limited.

- **Option B – SLA resin printing:** Fully dense, strong, watertight, precise O-ring grooves, but higher cost and more specialised equipment.

Selected: FDM (PLA) for prototype, SLA recommended for final version. FDM is adequate for short-term immersion (tested to 30min) and fits the budget but absorbs water over time. SLA would be used in a production version.

Sealing Method

- **Option A – Screw-on end caps:** Very secure but requires precise threading, difficult with FDM, risk of cross-threading.
- **Option B – Friction-fit with double O-rings:** Simple, self-aligning, proven in SCINI ROV [?]. O-rings provide friction and external water pressure increases seal force.

Selected: Friction-fit with double O-rings. This design allows easy assembly/disassembly (serviceability) and provides redundancy.

6.4.2 Sensor Design Choices

Chlorophyll Fluorometer – Detector

- **Option A – Discrete photodiode and optical filter:** BPW34 photodiode, red long-pass filter (e.g., Lee #105 Orange), transimpedance amplifier (LM358). Low component cost but filters are expensive and difficult to source.
- **Option B – Integrated multi-channel spectral sensor (AS7341):** 11 channels, on-chip interference filters (445nm and 680nm), programmable gain up to 256×, 16-bit ADC, I²C interface.

Selected: AS7341. It eliminates external filters, provides simultaneous excitation monitoring (445nm) and fluorescence (680nm) channels, and greatly simplifies implementation and testing.

Chlorophyll Fluorometer – Excitation LED

- **Option A – LUB32W (3mm blue LED):** 430nm peak, 79mcd luminous intensity, broad spectrum (400–460nm). Very dim.
- **Option B – CZ104BC (10mm blue LED):** 460nm peak, 5500–6000mcd, narrow beam angle (35°). Much brighter.

Selected: CZ104BC (R29). The LUB32W's low intensity would not excite measurable fluorescence whereas the CZ104BC provides bright excitation to aid in detecting low chlorophyll concentrations. It is also very close to the chlorophyll a excitation wavelength.

Temperature Sensor

- **Option A – Analog LM35:** 10mV/°C, requires ADC and noise filtering.
- **Option B – Digital DS18B20:** 1-Wire digital, factory calibrated, available in waterproof probe.

Selected: DS18B20 waterproof. Digital interface eliminates ADC noise, pre-waterproofed, meets Antarctic range.

Pressure/Depth Sensor

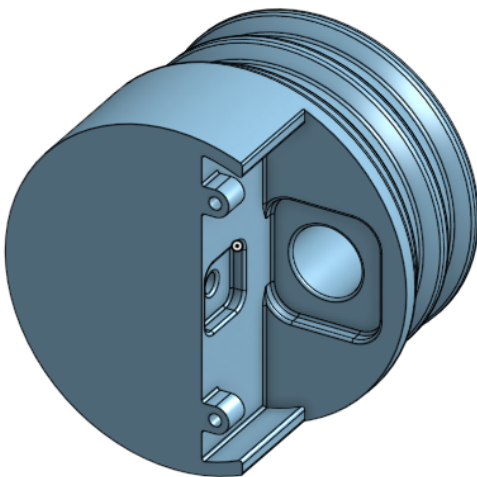
- **Option A – Digital I²C sensor (MS5837-30BA):** High accuracy, 300m range, but less available with longer lead time and higher cost.
- **Option B – Analog industrial sensor (0–1.2MPa, 0.5–4.5V):** Local stock at a lower price, IP65, 0–120m range, requires simple voltage divider.

Selected: Analog industrial sensor. Local availability and lower cost justify the slightly lower accuracy ($\pm 1.5\%$ FS) for surface profiling.

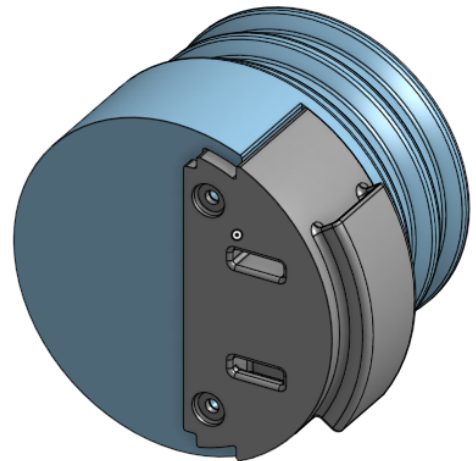
6.5 Final Design

6.5.1 Housing Fabrication

The main body is a 50mm OD PVC pipe (length 250mm, wall 1.6mm). The inner edge was chamfered for smooth O-ring assembly. Two end caps were FDM 3D-printed from PLA with 4 perimeters. One end cap houses the pressure and temperature sensors (epoxy-potted). The other end cap contains the AS7341 and 460nm LED, placed behind laser-cut clear acrylic windows (epoxied). The end cap includes a cover with openings designed to allow passive water flow while blocking ambient light. Both of the end caps have space to add sensors to provide a more complete sensor suite. A tether attachment point was glued to the pipe and lead ballast was inserted into the housing to ensure it sinks. The required ballast mass was calculated as $m_{\text{ballast}} = \rho_{\text{water}} V_{\text{disp}} - m_{\text{probe}} \approx 491 - 285 = 206 \text{ g}$ with an added margin, 250–300 g of lead is needed to ensure reliable sinking. This can be adjusted to get the desired characteristics.



(a) Fluorometer end cap without cover.



(b) External view with ambient light cover.

Figure 6.1: Fluorometer end cap design.

6.5.2 O-ring Groove Design

Double O-rings (46mm OD, 3mm cross-section) were used. Groove dimensions: width 3.2mm, depth 2.5mm (calculated for 17% compression), side walls had to be made at 45° which is a compromise to enable support-free FDM printing. The o-rings had to be stretched to 5% which is the recommended limit for male grooves in order to get the requires compression.

6.5.3 Sensor Integration

All sensors interface with an ESP32-C6:

- AS7341: I²C (SDA=GPIO21, SCL=GPIO22).
- DS18B20: 1-Wire (GPIO17) with 4.7k pull-up.
- Pressure sensor: analog output (0.5–4.5V) via voltage divider (R1=3.3k, R2=10k) to ADC (GPIO1).
- LED: driven by 2N2222 transistor (GPIO8, PWM, 5kHz).

Spare interfaces are available for future expansion (e.g., dissolved oxygen, conductivity).

6.5.4 Test Jig and Initial Validation

A dark enclosure with two ports (LED and AS7341 at 90°) was used to test spinach extract dilutions. The spinach was crushed in acetone and allowed to stabilise over 24hr before being filtered for use in testing. Dark and illuminated readings were taken and the difference (light–dark) gave net excitation (445nm) and fluorescence (680nm). Water-only showed only excitation scatter while chlorophyll samples showed clear fluorescence. The fluorescence/excitation ratio decreased with decreasing concentration after the first dilution showing promising results before moving on to a more complex sensing algorithm.

6.5.5 Fluorometer Timing Optimisation (Sensitivity Analysis)

The AS7341 integration time was varied:

- Fast: ATIME=3, ASTEP=500 → 5.6ms (many readings but low sensitivity).
- Balanced (selected): ATIME=10, ASTEP=500 → 15.3ms, gain 256× → 6–7 readings per 100ms, sufficient sensitivity.
- Slow: ATIME=100, ASTEP=999 → 280ms (too slow for ramp).

Step interval was set to 60ms (longer than 15.3ms + overhead). Ramp steps = 10, LED duty from 0 to 255.

6.5.6 Final Ramp Measurement Algorithm

On receiving the serial command R, the probe:

1. Reads temperature and depth once.

2. Takes dark reference (LED off, average of 10 readings).
3. Ramps LED brightness in 10 steps (60ms per step).
4. At each step, waits for integration (15.3ms + 20ms margin), then reads all 12 AS7341 channels.
5. Subtracts dark reference, outputs CSV row: `reading,time_ms,temp_c,depth_m,F1...F8,clear,NIR`.
6. Turns LED off and waits for next command.

Example output (water vs. chlorophyll) is shown in Table 6.4.

Table 6.4: Example fluorescence readings.

Sample	F2 (445nm)	F8 (680nm)	Ratio (F8/F2)
Water only	5394	260	0.0482
Chlorophyll (1/32 dilution)	5299	264	0.0498
Chlorophyll (full strength)	3507	296	0.0844

6.6 Testing and Results

The ATPs were executed with the results summarised in Table 6.5.

Table 6.5: ATP results for the sensor and housing subsystem.

ID	Test	Result
ATP01	Chlorophyll sensitivity	PASS – monotonic F8/F2 ratio increase down to 1/32 dilution
ATP02	Temperature accuracy	PASS – $\pm 0.4^{\circ}\text{C}$ vs reference
ATP03	Depth linearity	PASS – $R^2 > 0.99$ (0–1.5m)
ATP04	Total cost	PASS – Bill of materials R2000 (see Appendix)
ATP05	Hand-deployable	PASS – One person can carry and deploy
ATP06	Serviceability	PASS – End caps removable by hand; no leaks after reassembly
ATP07	Waterproof	PASS – No ingress at 1m for 30min
ATP08	Negative buoyancy	PASS – Probe sinks in water

All ATPs passed. The fluorometer reliably detects chlorophyll down to 1/32 dilution of spinach extract (relative concentration), temperature accuracy is within specification, depth measurements are linear, and the housing meets waterproofing and serviceability requirements.

6.7 Conclusion

This chapter presented the design, implementation, and testing of the sensor and housing subsystem for the Antarctic probe. The final design uses a PVC main tube with FDM-printed PLA end caps sealed by double O-rings (friction fit). The sensor suite comprises an AS7341 spectral sensor with a 460nm LED for chlorophyll fluorometry, a DS18B20 for temperature, and an analog pressure sensor for depth.

All sensors are integrated with an ESP32-C6 and communicate via a simple serial command (R) that triggers a 10-step LED ramp, producing CSV output ready for wireless transmission or local logging.

All acceptance test procedures passed, confirming that the probe meets the stakeholder's requirements for chlorophyll detection, temperature and depth measurement, hand-deployability, serviceability, waterproofing, and negative buoyancy, all within the R2000 budget. Limitations (FDM end caps not suitable for long-term immersion, relative chlorophyll calibration, pressure sensor temperature rating) are noted, and future improvements (SLA printing, absolute calibration, wider-range pressure sensor) are recommended.

Chapter 7

Conclusions

The purpose of this project was to design and demonstrate a bipartite marine environmental monitoring system capable of continuously gathering microbial proxy data in the Antarctic region without requiring a human operator in the field. The system was developed in direct response to the requirements of Dr Emma Rocke (MARiS, UCT), whose research into marine microbial ecosystems is presently constrained by the high cost, infrequency, and physical danger of conventional Antarctic research cruises.

The report began with a comprehensive literature review (Chapter ??), which established the engineering constraints governing each subsystem. The review confirmed that conventional radio-frequency communication cannot bridge the water-air boundary, validated the store-and-forward deployment architecture, identified Lithium Thionyl Chloride and Ferroelectric RAM as the preferred storage and energy technologies for sub-zero operation, and justified an optical sensing approach centred on chlorophyll-a fluorometry as the most field-ready microbial proxy available within the project budget.

The four subsystem chapters then detailed the design, implementation, and acceptance testing of each engineering component.

The Data Transfer and Visualisation subsystem (Chapter ??) evaluated three wireless protocols — NFC, BLE, and ESP-NOW — against figures of merit including range, power consumption, implementation cost, and hardware reliability. NFC was ruled out due to strict proximity and orientation requirements unsuitable for use on the open ocean. BLE, despite a complete firmware implementation including a full GATT profile and CRC-16 packet validation, was abandoned after the ESP32-C6's BLE advertising proved undetectable on both available development boards, a failure attributed to immaturity in the RISC-V BLE stack at the time of development. ESP-NOW was selected on the basis of proven hardware reliability, ultra-low power consumption, and a 100 m operational range far exceeding the 10 m acceptance criterion. The Mission Control GUI, implemented as a fully offline Python application, passed all seven acceptance test procedures, including a 1.2% average data error margin against the 5% target and an 8 s average wake-to-advertising latency within the 10 s specification.

The Edge Architecture and Explainable AI subsystem (Chapter ??) implemented a deterministic four-state firmware lifecycle on the ESP32-C6 to prevent surface-level data pollution, with RTC SRAM providing non-volatile record persistence across deep sleep cycles. An Isolation Forest augmented with a deterministic Explainable AI ruleset — calibrated against physical spectral measurements of the AS7341 sensor — achieved 92% anomaly classification accuracy with zero false positives, executing in approximately 0.008 s per record against a 2 s specification.

The Power and Thermal Subsystem (Chapter ??) selected an ER14505 Lithium Thionyl Chloride battery stack and an MCP1700 ultra-low-quiescent LDO regulator. Energy modelling demonstrated approximately 22 measurement cycles per day over a 365-day mission, validating the annual autonomy requirement. A passive multi-layer thermal insulation strategy maintained the internal core at -12°C after six hours in a -30°C cold-soak, exceeding the 15°C differential specification. Measured standby current of $16.8\mu\text{A}$ confirmed compliance with the $20\mu\text{A}$ sleep budget.

The Sensors and Housing subsystem (Chapter ??) produced a 50 mm OD PVC probe housing sealed by friction-fit double O-rings, integrating an AS7341 spectral sensor with a 460 nm LED fluorometer, a DS18B20 temperature sensor, and an analog pressure sensor for depth. All eight acceptance test procedures passed, including a monotonic fluorescence-to-excitation ratio response down to 1/32 chlorophyll dilution, temperature accuracy of $\pm 0.4^{\circ}\text{C}$, depth linearity of $R^2 > 0.99$, and confirmed waterproofing at 1 m depth for 30 minutes — all within the R2000 budget constraint.

In summary, the project achieved its stated objectives. A functional, fully integrated bipartite monitoring system was designed and demonstrated across all four subsystems. The probe acquires, stores, and wirelessly offloads optical and physical sensor data to a laptop-based Mission Control interface that processes and visualises the measurements with localised machine learning anomaly detection, operating entirely offline and without any physical port on the sealed enclosure. The system is hand-deployable by a single non-specialist operator from a vessel of opportunity, directly addressing the safety and logistical constraints identified by the client. Remaining limitations — including FDM end caps not suitable for long-term immersion, the absence of an absolute chlorophyll calibration, and the pending maturation of the ESP32-C6 BLE stack — are documented in Chapter 8 with targeted paths to resolution in future design iterations.

Chapter 8

Recommendations

This chapter consolidates recommendations for future development of the Antarctic Microbial Sensor system, drawing on the limitations identified during testing and the constraints imposed by the project timeline and budget. The recommendations are organised by subsystem and followed by system-level directions that cut across the architecture as a whole.

8.1 Data Transfer and Visualisation

Migration to BLE

The most impactful near-term improvement to the data transfer subsystem would be to revisit Bluetooth Low Energy once the ESP32-C6 NimBLE stack matures. At the time of development, BLE advertising was undetectable on both available development boards despite a complete firmware implementation, a failure attributed to immaturity in the RISC-V BLE stack rather than a design flaw. Migrating to BLE would eliminate the requirement for a separate receiver dongle, reducing system cost by approximately R230, simplifying field deployment, and consolidating the communication architecture to a single chip already present on the probe. The full GATT profile — METADATA, CONTROL, DATA, and STATUS characteristics — has already been implemented and validated in simulation; the migration path is therefore low-risk once hardware-level advertising is confirmed working.

Encryption of the ESP-NOW Link

The current ESP-NOW data transfer is unencrypted. Although the protocol operates in a point-to-point mode that limits interception risk in open-ocean deployments, adding the built-in CCMP (AES-128) encryption layer supported natively by ESP-NOW would protect the integrity of scientific data against interception in shared RF environments such as busy harbour berths or research stations. This is a configuration-level change requiring no additional hardware.

Satellite Uplink Integration

The current architecture terminates at the researcher’s laptop. A natural extension would be to integrate a low-power satellite modem (such as a RockBLOCK 9603 Iridium module) into the docking station or laptop-side software to relay high-value anomaly alerts to mainland researchers over the constrained Antarctic satellite link. This would complete the end-to-end data pipeline envisioned in

the original system design and directly addresses the stakeholder’s need for real-time environmental warnings from the field.

8.2 Edge Architecture and Explainable AI

Physical F-RAM Integration

The current implementation simulates the Ferroelectric RAM (F-RAM) hardware abstraction layer using the ESP32-C6’s internal RTC SRAM, which was mathematically sound for proof-of-concept validation but is limited in capacity and would not survive a complete power loss. Future iterations should integrate a dedicated SPI-based F-RAM module, such as the Fujitsu MB85RS series, which offers 10^{14} read/write endurance cycles, sub-500 μA write current, and reliable data retention down to -55°C . This component is a direct drop-in once the SPI bus is available, and its cost is modest relative to the reliability improvement it provides for a twelve-month autonomous mission.

Turbidity-Aware XAI Calibration

The Explainable AI ruleset was calibrated against bench measurements taken in relatively clear aquatic media. In highly turbid Antarctic waters, wavelength-dependent light attenuation governed by Beer-Lambert’s Law will cause fluorescence ratios to deviate from the calibrated thresholds in ways that could produce false-positive biological classifications. Future iterations should incorporate a real-time turbidity correction step, dynamically adjusting the F8/F2, F5–F6/F2, and F1–F3/F2 threshold constants as a function of ambient turbidity measured by the clear and NIR channels of the AS7341. This would make the XAI engine robust to the particle-heavy meltwater conditions characteristic of Antarctic spring blooms.

Expanded Training Dataset

The Isolation Forest model was trained and validated on a synthetic dataset. Collaborating with Dr Roche’s group to obtain labelled field data from existing Antarctic deployments would allow the model to be retrained on real spectral signatures, improving classification accuracy beyond the current 92% and reducing the risk of spurious anomaly flags under genuine operational conditions.

8.3 Power and Thermal Subsystem

Active Thermal Management

The passive multi-layer insulation strategy maintained the internal core at -12°C after a six-hour cold-soak at -30°C , satisfying the 15°C differential specification. However, the Antarctic winter can sustain ambient temperatures below -50°C for extended periods that exceed the duration of the bench test. A resistive heating element — thermally bonded to the battery stack and duty-cycled by the MCU when core temperature falls below a threshold — would provide active thermal regulation and extend the reliable operating envelope to the full rated temperature range of the ER14505 cell.

Battery Stack Scalability

The energy model demonstrated approximately 22 measurement cycles per day with the current 2400 mAh battery stack derated to 1200 mAh at -50°C . Increasing the battery stack to two cells in parallel would double the usable capacity, raising the daily cycle count to approximately 44 and enabling sub-hourly sampling intervals. This is particularly relevant for detecting rapid-onset harmful algal bloom events, which can develop over timescales of hours. The split-rail power architecture and PCB layout should be sized to accommodate a two-cell stack without requiring a redesign of the regulator circuit.

PCB Implementation

All power circuitry was assembled and validated on a breadboard prototype. Migrating to a custom PCB would reduce parasitic resistance on the high-current battery and sensor rails, improve thermal bonding between the MCP1700 regulators and the battery stack, and allow a more compact form factor. Minimum trace widths of 0.5 mm on the switched 5 V rail are sufficient to handle the peak sensor current, as established during schematic validation.

8.4 Sensors and Housing

SLA End Caps for Long-Term Deployment

The FDM-printed PLA end caps were adequate for the 30-minute waterproofing acceptance test but are not suitable for prolonged immersion. PLA absorbs water over time, causing dimensional changes that compromise O-ring sealing. Transitioning to Stereolithography (SLA) resin end caps for a production version would yield fully dense, watertight parts with the dimensional accuracy required for reliable O-ring groove geometry across repeated deployment cycles.

Absolute Chlorophyll Calibration

The fluorometer currently reports relative fluorescence-to-excitation ratios, which are sufficient for detecting community shifts but cannot provide absolute chlorophyll-a concentrations in $\mu\text{g/L}$ without a calibration curve. Generating this curve requires measurements against certified chlorophyll standards or co-deployment with a calibrated reference instrument. Establishing an absolute calibration would significantly increase the scientific value of the probe's output and make it directly comparable to data from commercially validated instruments such as the Aanderaa Chl-a sensor.

Expanded Sensor Suite

The current design provides chlorophyll-a, temperature, and depth. Dr. Rocke identified dissolved oxygen, salinity, and nutrients as additional parameters of scientific interest that were excluded due to budget constraints. Both end caps were designed with spare sensor ports and the ESP32-C6 has unused I2C and ADC GPIO available. Adding a phase-shift oxygen optode (e.g. an Aanderaa 4330F-compatible sensor) and a conductivity cell in a future revision would provide a full CTD+O₂+Chl-a profile, substantially increasing the ecological value of each deployment.

Deeper-Water Pressure Housing

The current PVC housing is validated to 1 m depth for the prototype. Scaling to operationally relevant depths of 50–200 m would require transitioning the main body to acrylic or aluminium tube, upgrading the O-ring groove geometry to handle the increased hydrostatic load, and replacing the analog pressure sensor (rated to 1.2 MPa, 120 m) with the MS5837-30BA digital sensor rated to 300 m. This path is architecturally straightforward given the modular end cap design.

8.5 System-Level Recommendations

Field Validation in Representative Conditions

All testing was conducted in controlled laboratory and pool environments. Prior to operational Antarctic deployment, the integrated system should undergo field validation in open coastal waters under representative temperature and wave conditions. This would expose coupling effects between subsystems — for example, the impact of probe motion on the fluorometer’s synchronous detection and the effect of wave action on ESP-NOW link reliability — that cannot be characterised in the laboratory.

Autonomous Multi-Dive Sequencing

The current architecture requires a researcher to connect the GUI and issue a PREPARE command before each dive. Adding a scheduled autonomous dive mode — in which the probe descends, logs, ascends, and signals readiness for data offload on a configurable timer — would reduce operator interaction and enable deployment on unmanned vessels of opportunity where no researcher is continuously present.

Standardised Data Format for Interoperability

The raw CSV output from the probe is processed exclusively by the Mission Control GUI. Adopting a community-standard format such as the Ocean Data Standards Consortium’s NetCDF or the Argo float standard binary profile format would allow the probe’s output to be ingested directly by existing oceanographic data repositories and compared against the broader Argo float network dataset, increasing the scientific return from each deployment with minimal additional engineering effort.

Chapter 9

Appendix

Appendix A

Dev Mode: Simulated Data Screenshots

The following screenshots were captured during Dev Mode: Simulate sessions and demonstrate the GUI's data display and graph capabilities independently of a physical probe. Note that the simulated values are synthetically generated and are not intended to represent realistic oceanographic measurements; their purpose is solely to validate the rendering, scrolling, graphing, group navigation, and anomaly flagging behaviour of the GUI pipeline.

Dataset A — 28 Records (No Anomalies)

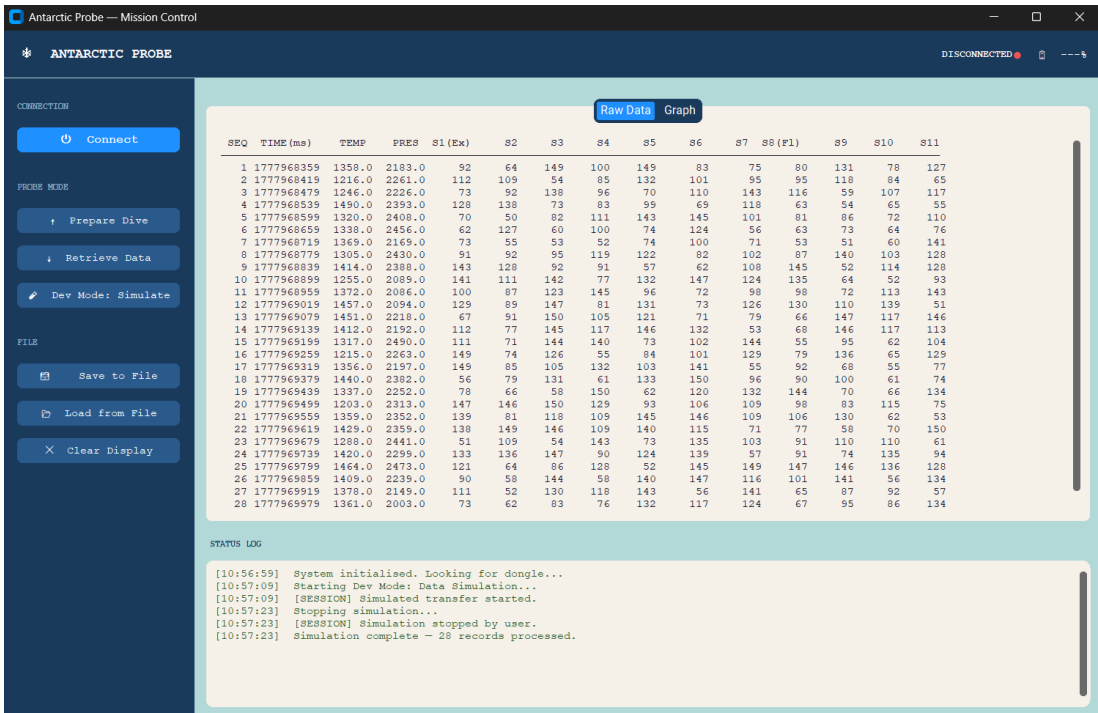


Figure A.1: Raw Data tab displaying 28 simulated records received via Dev Mode: Simulate

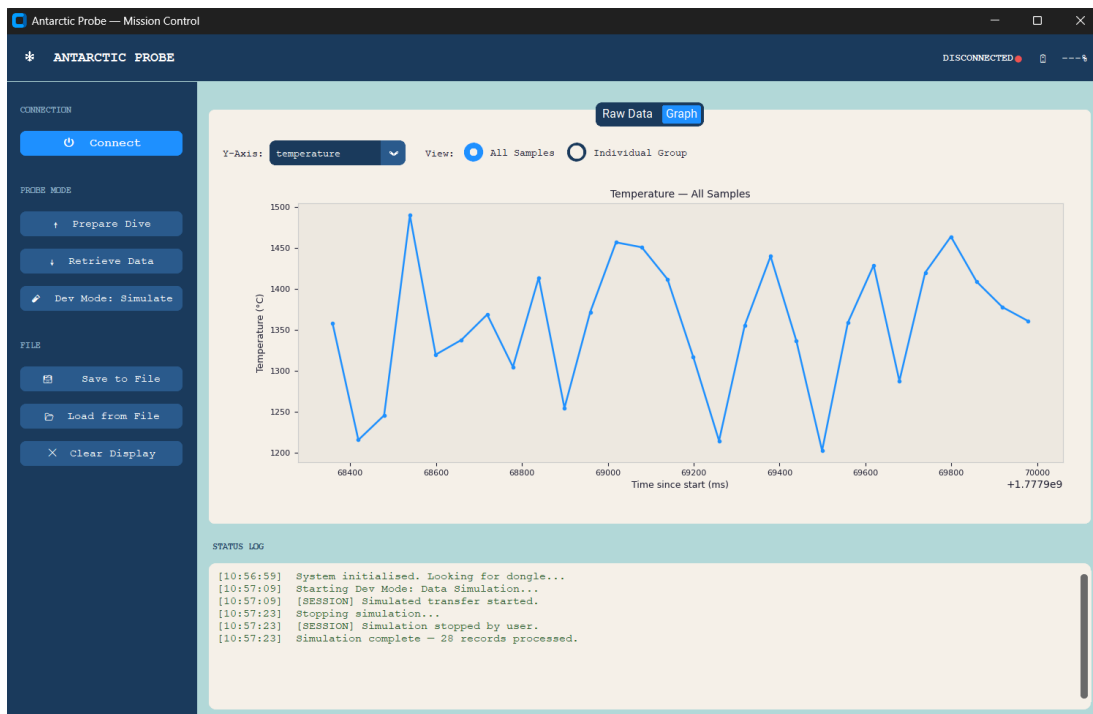


Figure A.2: Graph tab — All Samples view of temperature over time for 28 simulated records

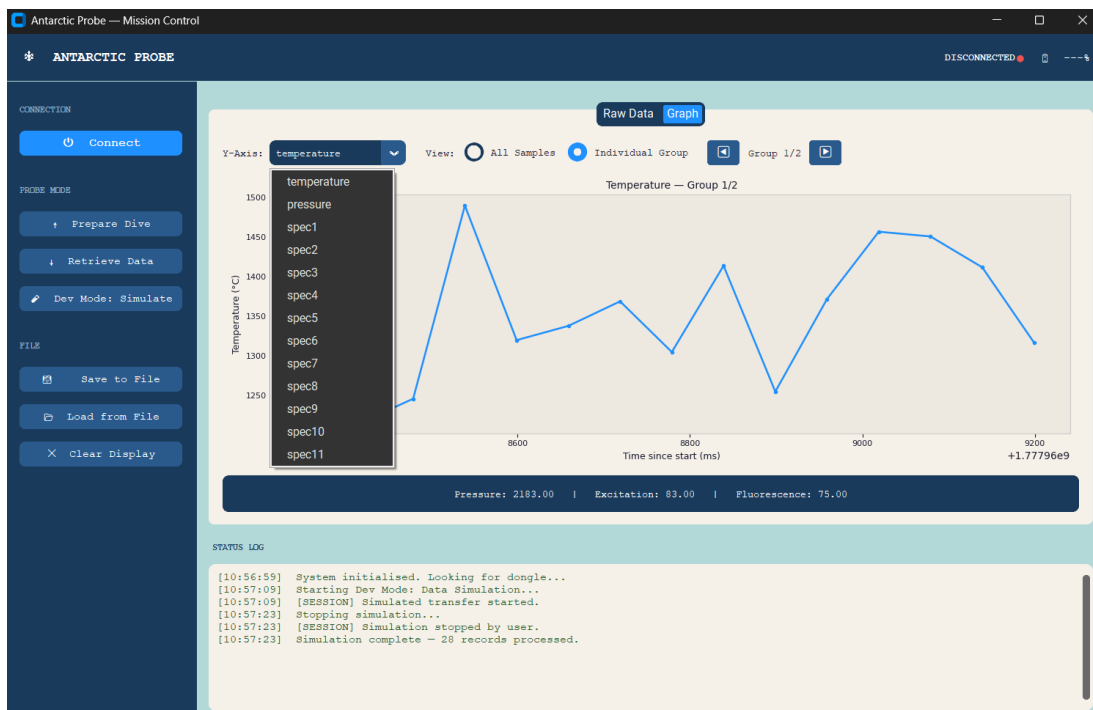


Figure A.3: Graph tab — Individual Group view (Group 1/2) with the *y*-axis dropdown open, showing all selectable sensor channels

Dataset B — 113 Records (With Anomalies)

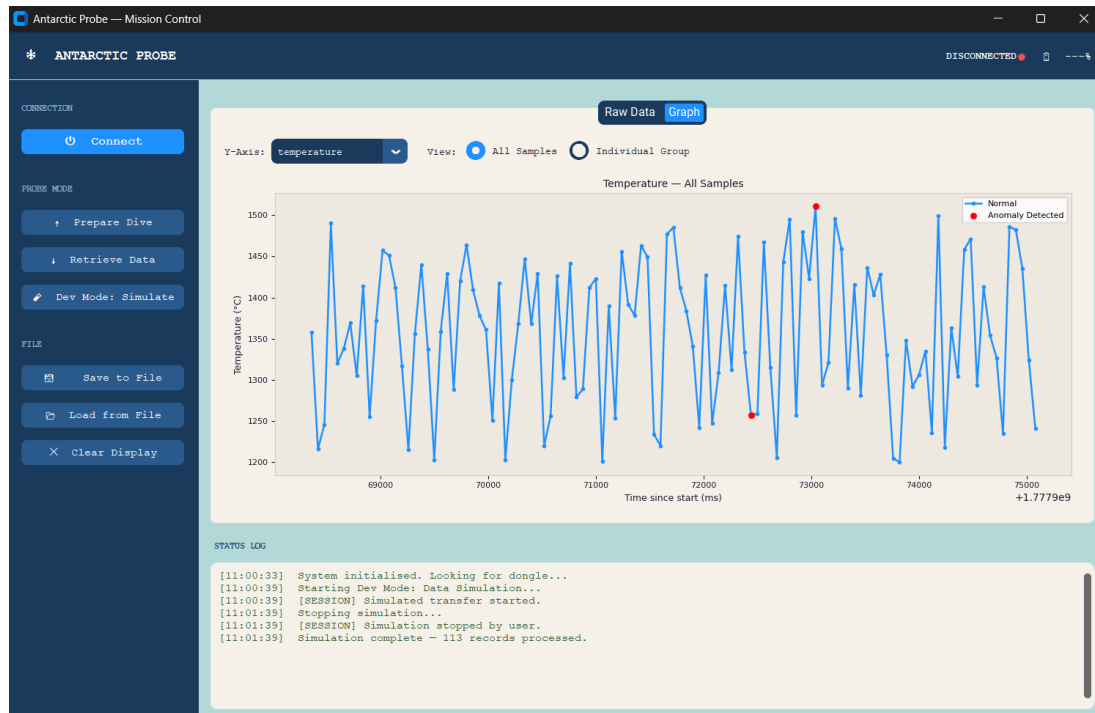


Figure A.4: Graph tab — All Samples view of temperature over time for 113 simulated records. Red markers indicate records flagged as anomalous by the `MarineAnomalyDetector`. The status log confirms 113 records were processed.

Appendix B

Evidence of GA

B.1 Mogamad Saeed Solomon - SLMMOG032

Table B.1: Evidence of GAs for SLMMOG032

GA	Evidence met
GA3	Refer to subsystem 1, Data transfer and visualization
GA7	
GA8	Refer to the Microsoft teams channel chat tab
GA10	Refer to the shared documents tab on the Microsoft teams channel where all meeting minutes are uploaded

B.2 Joshua Naidoo - NDXJOS012

Table B.2: Evidence of GAs for NDXJOS012

GA	Evidence met
GA3	Refer to Power and Thermal Subsystem
GA7	
GA8	Refer to the Microsoft teams channel chat tab
GA10	Refer to the shared documents tab on the Microsoft teams channel where all meeting minutes are uploaded

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